

Diabetic Kidney Disease

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KEY POINTS

- The total number of people worldwide with diabetic kidney disease (DKD), or chronic kidney disease with diabetes, continues to rise in parallel with the rapidly increasing prevalence of type 2 diabetes.
- Multiple risk factors are associated with the development of DKD including, but not limited to, suboptimal glycemic control, hypertension, dyslipidemia, obesity, smoking, intrauterine factors, epigenetic and genetic factors, and dietary patterns.
- Optimized glycemic control, blood pressure management, lipid management, and use of organ protective therapies constitute evidence-based treatment of patients with DKD.

Diabetic kidney disease (DKD) is defined as chronic kidney disease (CKD) attributable to diabetes mellitus. CKD is determined by sustained elevation of urinary albumin excretion (urine albumin-to-creatinine ratio [UACR] >30 mg/g) or a reduction in estimated glomerular filtration rate (eGFR) to <60 mL/min/1.73 m².¹⁻³ Approximately 90% of the 537 million adults aged 20 to 79 years with diabetes mellitus have type 2 diabetes, and 9 out of 10 live in low- and middle-income countries.⁴ Since DKD develops in about one-third of patients with type 1 diabetes and up to half of patients with type 2 diabetes, the disproportionate burden of the increase in DKD (63%) has occurred in lower-middle-income countries.⁴⁻⁷ As a leading cause of kidney failure and the most frequent indication for kidney replacement therapy (KRT), this diabetes complication is a major public health problem worldwide.⁸⁻¹⁰ Individuals with CKD are more likely to die from cardiovascular disease, particularly heart failure and atherosclerotic disease, than progress to kidney failure. Furthermore, they are at remarkably high risk of infections, hospitalizations, and related deaths.¹¹⁻¹⁵ The prevalence of kidney failure requiring KRT in the United States doubled between the years 2000 and 2019, with diabetes as the leading cause.^{16,17} Because of limited access to KRT in many regions of the world, it is estimated that the

number of people who die prematurely because of lack of KRT access is up to three times higher than the number who actually receive treatment.^{10,18-20} Consequently, during the time period spanning 1990 to 2016, the death rate due to CKD increased globally by 98% (Fig. 41.1).^{21,22}

Efforts to improve prevention, detection, and intervention for DKD are critical given the steadily increasing cases and availability of highly effective kidney and heart-protective agents. This chapter outlines the epidemiology, complex pathophysiology, risk factors, and natural history of DKD. It offers an overview of recommendations for DKD diagnosis, classification, use of guideline-directed medical therapies, and multidisciplinary, coordinated care strategies.

EPIDEMIOLOGY OF DIABETIC KIDNEY DISEASE

In parallel to the observed increased prevalence of diabetes mellitus in the United States, the prevalence of DKD ranges between 26% and 44%, with the total burden of CKD increasing by 53%, driven primarily by diabetes.^{21,23} Similar to the increase in CKD cases reported in the United States, the global prevalence of CKD in the 1990 to 2016 time period

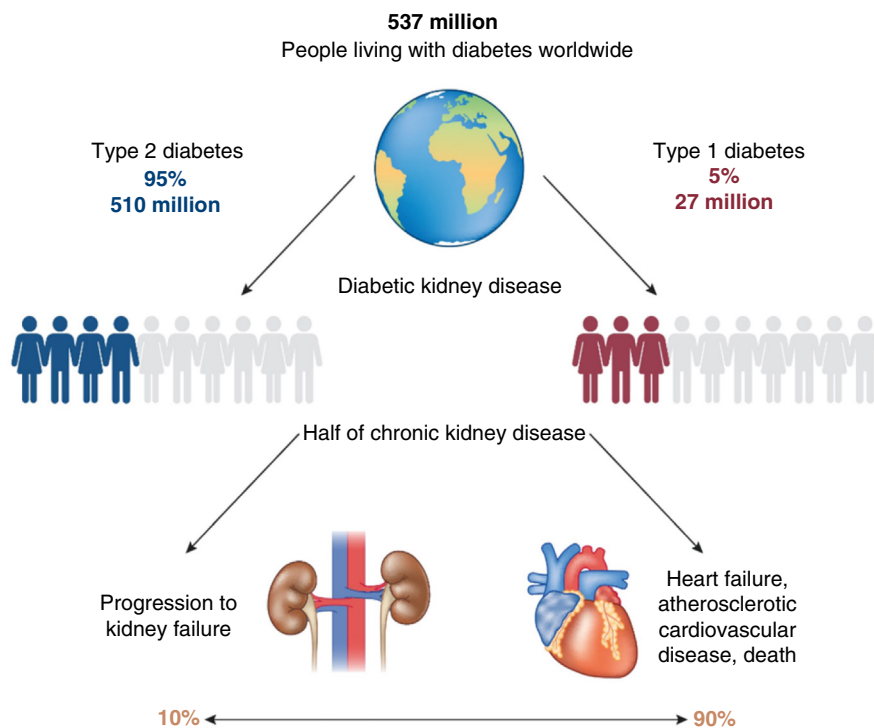


Fig. 41.1 Worldwide prevalence of diabetes.²² The worldwide prevalence of diabetes was estimated at 537 million people in the year 2021. Approximately 40% of people with type 2 diabetes (95% of all diabetes cases) and 30% of those with type 1 diabetes (5% of all diabetes cases) develop diabetic kidney disease. Half of all cases of CKD are attributable to diabetes. Most people with diabetic kidney disease (90%) die, mainly of heart failure or atherosclerotic cardiovascular events, with only a minority (10%) surviving to reach kidney failure.

increased by 87%, which has been attributed to an increase in the age-standardized prevalence of DKD.⁷ The United Kingdom Prospective Diabetes Study (UKPDS), which enrolled newly diagnosed participants with type 2 diabetes, offered an opportunity to assess incident DKD. After a median follow-up of 15 years, 38% of participants developed albuminuria, 29% developed creatinine clearance <60 mL/min or doubling of plasma creatinine, and 14% developed both conditions.⁶ Another seminal study, the Diabetes Control and Complications Trial (DCCT) demonstrated that 12% of participants developed microalbuminuria (albumin-to-creatinine ratio [ACR] >40 mg/24 hours) and 2.7% developed macroalbuminuria (ACR ≥ 300 mg/24 hours) after a mean follow-up of 6.5 years.²⁴ After 16 years of follow-up, the cumulative incidence of DKD defined as eGFR <60 mL/min/1.73 m² or kidney failure requiring dialysis or kidney transplantation in type 1 diabetes was 31.7%.²⁵ A meta-analysis of 71 studies of both types of diabetes reported the annual incidence of albuminuria at 8% in type 2 diabetes and 2% to 3% in type 1 diabetes, while the new-onset eGFR <60 mL/min/1.73 m² occurred at 2% to 4% per year regardless of diabetes type.²⁶ A contemporary study of secular trends has shown recent declines in CKD incidence among persons with diabetes in the Center for Kidney Disease Research, Education, and Hope (CURE-CKD) registry.²⁷ CKD incidence rates standardized by age, race and ethnicity, and sex were 81.6 (78.0–85.2)/1000 person-years during the 2015–2016 time period, 73.7 (71.5–75.8)/1000 person-years in 2017–2018, and 64.0 (62.2–65.9)/1000 person-years in 2019–2020. Across these time periods, CKD incidence was substantially higher among American Indian/Alaska Native, Black, Hispanic/Latino(a), and Native Hawaiian/

Pacific Islander groups and lower in Asian persons when compared with a White reference group (Fig. 41.2).²⁷ In the context of declining CKD incidence, the higher prevalence rates of DKD and kidney failure may reflect reduced rates of cardiovascular events and deaths, thus allowing greater numbers of patients to survive to kidney failure in recent years.²⁸

Clinical Relevance: Epidemiology

The prevalence of CKD continues to rise in parallel with the increased prevalence of diabetes globally. Importantly, incidence rates of CKD are substantially higher in American Indian/Alaska Native, Black, Hispanic/Latino(a), and Native Hawaiian/Pacific islander groups when compared with White persons.

DIAGNOSIS AND CLASSIFICATION

DKD refers to a diagnosis of CKD in a patient with diabetes without other known or likely causes. The absence of indicators of other causes of CKD or presence of diabetic retinopathy is highly suggestive of a DKD diagnosis.^{2,29,30} Since the onset of type 2 diabetes is often insidious, duration of diabetes in type 2 diabetes is known with less certainty than it is for type 1 diabetes. For this reason, screening for DKD in type 2 diabetes is recommended from the time of diabetes diagnosis, whereas screening is recommended only after 5 years of type 1 diabetes unless the patient has hypertension.^{2,29-32} The preferred test for albuminuria is the UCAR performed on a random urine spot sample, if

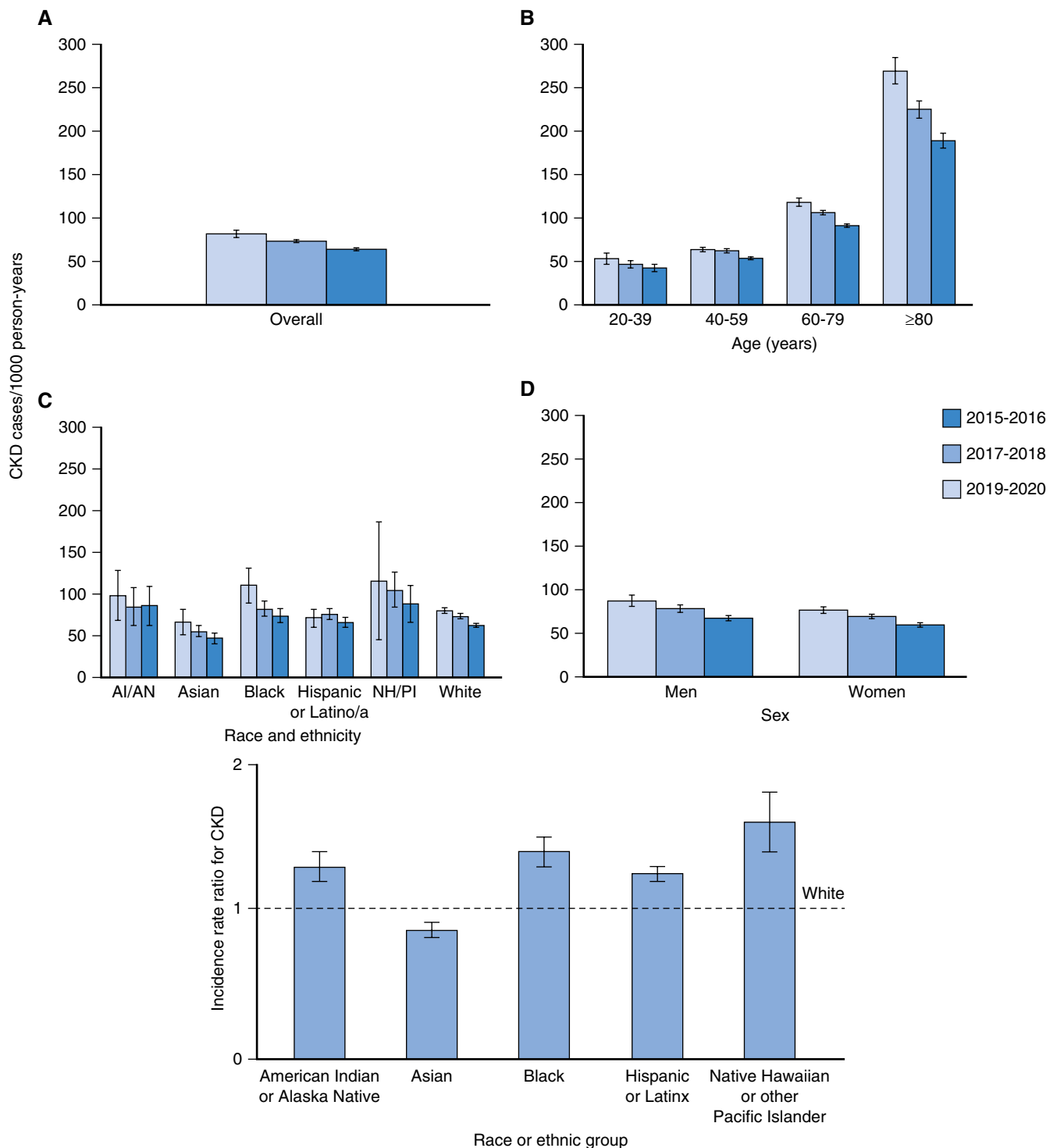


Fig. 41.2 Incidence rate ratios for chronic kidney disease among patients with diabetes.²⁷ Panel 1: (A) Chronic kidney disease (CKD) incidence with 95% confidence intervals (CIs) in diabetes over time. Overall CKD incidence standardized to the 2010 U.S. Census Bureau population by age, race and ethnicity, and sex. (B) Age-specific, adjusted for sex, race, and ethnicity. (C) Race and ethnicity specific, adjusted for sex and age. (D) Sex specific, adjusted for age, race, and ethnicity. AI/AN, American Indian/Alaskan Native; NH/PI, Native Hawaiian/Pacific Islander). Panel 2: Incidence rate ratios for CKD among patients with diabetes, stratified according to race and ethnic group, between 2015 and 2020. The dashed line represents the incidence rate ratio among White patients (reference group). The analysis was adjusted for age and sex. I bars indicate 95% CIs.

possible, in the morning. The eGFR is calculated from the serum creatinine concentration using either the Modification of Diet in Renal Disease or preferably the updated Chronic Kidney Disease-Epidemiologic Prognosis

Initiative (CKD-EPI) equation. The latter evidence-based equation no longer includes a parameter that adjusts for differences in creatinine production based on continental ancestry (race-free eGFR estimation equation). However,

in some situations where precise estimation of GFR might be required for medication dosing or other individual clinical decisions, the serum marker cystatin C, which is not affected by variables that influence creatinine production, can be entered either alone or combined with the serum creatinine level in the updated CKD-EPI equation to achieve even greater precision for eGFR.³³⁻³⁶ However, since cystatin C is not standardized or available in many clinical laboratories, major efforts are under way to improve its affordable availability and standardized values.

UACR and eGFR should be measured more frequently (every 3–6 months) with rapid progression of CKD (e.g., loss of ≥ 5 mL/min/1.73 m² in eGFR per year) or following initiation of kidney-heart protective medications, including angiotensin-converting enzyme (ACE) inhibitors, angiotensin-receptor blockers (ARB), sodium-glucose cotransporter 2 (SGLT2) inhibitors, glucagon-like peptide-1 (GLP-1) agonists, or aldosterone blockade.^{2,29,30,37} When used together, eGFR and UACR improve risk stratification and diagnostic accuracy (Fig. 41.3).^{3,38} Notably, the presence of nondiabetic CKD in patients with diabetes ranges between 25% to up to 50% of cases based on biopsy studies.³⁹⁻⁴² Therefore evaluation at the time of diagnosis should include a comprehensive medical history to inquire about the presence of other systemic conditions with possible kidney involvement (e.g.,

autoimmune disease, chronic infections, malignancies, and frequent urinary tract infections), a careful family history (e.g., history of polycystic kidney disease), review of social and environmental factor exposure, and a comprehensive physical examination.⁴²⁻⁴⁴ In routine clinical practice, diagnostic kidney biopsies are performed infrequently in persons with diabetes and CKD, mostly if there is a clinical suspicion for an alternative diagnosis suggested by atypical features like sudden onset of proteinuria, change in clinical status (rapid CKD progression), presence of hematuria, or shorter duration of diabetes (Table 41.1).^{3,29,30,37,39,41} Histomorphologic changes identified through kidney biopsy may include findings consistent with DKD, nondiabetic CKD, or nondiabetic CKD superimposed on DKD. Nondiabetic CKD changes encompass a broad spectrum of diagnoses, including primary glomerular diseases (e.g., focal glomerular segmental glomerulosclerosis, IgA nephropathy, and membranous glomerulonephritis), hypertension with nephrosclerosis, and acute tubular injury.^{40,41,45-49} Importantly, patients with structural changes indicative of DKD are at higher risk for kidney failure when compared with patients with nondiabetic forms of CKD. For instance, in a prospective study of 2484 patients with diabetes and CKD who had a kidney biopsy performed, after a median follow-up of 4.4 years, patients with DKD and eGFR 30 to 44 mL/min/1.73 m² had a higher risk for progression to kidney failure

Prognosis of CKD by GFR and albuminuria categories: KDIGO 2012				Persistent albuminuria categories Description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30 mg/g <3 mg/mmol	30-300 mg/g 3-30 mg/mmol	>300 mg/g >30 mg/mmol
GFR categories (mL/min per 1.73 m ²) Description and range	G1	Normal or high	≥ 90			
	G2	Mildly decreased	60-89			
	G3a	Mildly to moderately decreased	45-59			
	G3b	Moderately to severely decreased	30-44			
	G4	Severely decreased	15-29			
	G5	Kidney failure	<5			

Green: low risk (if no other markers of kidney disease, no CKD); yellow: moderately increased risk; orange: high risk; red, very high risk.

Fig. 41.3 Categorization and prognosis of chronic kidney disease.³

Table 41.1 Indications for Kidney Biopsy^{39,41}

Diabetic disease duration <5 years (in type 1 diabetes)
Absence of diabetic retinopathy (particularly in type 1 diabetes)
Rapid onset and progression of albuminuria or sudden onset of nephrotic syndrome
Rapid decline of kidney function (GFR) with or without albuminuria (>5 mL/min/1.73 m ² or ≥25% of baseline eGFR)
Presence of hematuria (dysmorphic erythrocytes) or active urine sediment
Clinical suspicion of other nephropathies (e.g., vasculitis and amyloidosis), or positivity for markers of other systemic diseases (e.g., ANCA, ANA, dsDNA, low complement factors, and cryoglobulinemia).
Prognostic evaluation in selected patients with clinical suspicion of DN.

ANA, Antinuclear antibodies; ANCA, antineutrophil cytoplasmic antibody; DN, diabetic nephropathy; dsDNA, double-stranded DNA; GFR, glomerular filtration rate.

compared with those with other forms of CKD. Hazard ratios (95% confidence intervals for progression to kidney failure) were 7.1 (2.46–20.49) and 0.89 (0.19–4.24) for patients with DKD and nondiabetic CKD, respectively.⁵⁰

RISK FACTORS

Numerous risk factors contribute to the development and progression of DKD. Hypertension and hyperglycemia can be modified through conventional approaches to risk factor control with medications and lifestyle modification. Smoking and healthy dietary patterns are also important goals for lifestyle modification. Inherited risk factors cannot be changed, but modification of epigenetic factors that influence gene expression may reduce risk of DKD, despite genetic susceptibility. In this section, we review risk factors that contribute to the development and progression of DKD. In addition to traditional risk factors for DKD, we included others that might be relevant in subsets of individuals experiencing uncommon exposures.

HYPERGLYCEMIA

Increased blood glucose concentration is the clinical hallmark of diabetes and is a cardinal determinant of its complications. Given the prominent role of hyperglycemia in the development of diabetic complications, it is not surprising that the duration of diabetes is one of the most important risk factors for DKD, and its influence is far greater than that of age, sex, or type of diabetes.^{6,38,51-53} Furthermore, most of the large observational studies indicate that hyperglycemia is an important predictor of progression to advanced CKD in both type 1 and type 2 diabetes.⁵⁴⁻⁵⁹ For instance, in type 1 diabetes, poor glycemic control independently predicts progression to development of albuminuria and kidney failure, and in type 2 diabetes higher hemoglobin A1c (HbA1c) levels are one of the

strongest determinants of DKD progression independent of albuminuria.⁶⁰⁻⁶²

Early evidence for the role of hyperglycemia in the development of DKD comes from biopsy studies in identical twins discordant for type 1 diabetes and from morphologic studies before and after pancreas transplantation.^{63,64} Glomerular changes including widened glomerular and tubular basement membranes and increased mesangial fraction are identified only in the diabetic member of twin pairs, suggesting that metabolic status, and not genetic predisposition, is responsible for the development of these diabetic kidney lesions.⁶³ In addition, prolonged normoglycemia following pancreas transplant in persons with type 1 diabetes and established DKD promotes virtually complete reversal of glomerular and tubular basement membrane thickness and increases in mesangial and interstitial volumes.⁶⁴ Nevertheless, the relative contributions of hyperglycemia, insulin resistance (IR), and alterations in endogenous insulin production to the development and progression of DKD remain unclear.

Further evidence that supports the role of early control of hyperglycemia comes from the DCCT/ Epidemiology of Diabetes Interventions and Complications (EDIC) study and UKPDS.⁶⁵⁻⁶⁷ Both studies demonstrated a beneficial effect of more rigorous glycemic control early in the course of diabetes, using well-delineated sets of glucose-lowering medications and interventions available at the time of the studies, respectively, for both T1DM (EDIC) and T2DM (UKPDS). Remarkably, the greater preservation of GFR and reduction in DKD in the intervention groups in both studies remained statistically significant despite dissipation and loss through much of the subsequent duration of the studies of the impressive glycemic control differences (with HbA1c as one of the important measures of metabolic control) achieved in the early years after enrollment. In light of these results, it was hypothesized that the observed beneficial effect of early strict glycemic control, termed the “legacy effect” or “metabolic memory,” could be explained by prevention of metabolism-driven epigenetic modifications.⁶⁸ The mechanistic role of epigenetic alterations in DKD is supported by the observation that periods of uncontrolled hyperglycemia are associated with an elevated risk of developing DKD that does not reflect real-time HbA1c levels.^{69,70} The “legacy effect” may, at least in part, explain discrepant results from different observational studies evaluating the predictive value of HbA1c for DKD.

Persistent hyperglycemia causes dysregulation of several effector molecules through various biochemical pathways in the kidney, including generation and accumulation of advanced glycation end products, increased activity of the polyol pathway, and activation of vasoactive hormones, such as angiotensin II and endothelin.⁷¹⁻⁷³ Activated prosclerotic cytokines, such as transforming growth factor beta (TGF-β) and vascular endothelial growth factor (VEGF), are important mediators between the metabolic and hemodynamic pathways affected by hyperglycemia that ultimately lead to the pathologic changes of DKD.⁷¹

INSULIN RESISTANCE

A cross-sectional study of 100 patients with type 2 diabetes found that albuminuric patients had lower glucose disposal

rates than nonalbuminuric patients.⁷⁴ The clinical evidence that IR precedes DKD came from prospective studies in the early 1990s. A prospective analysis of 108 patients with type 2 diabetes and normoalbuminuria who underwent euglycemic clamps at enrollment demonstrated that hyperinsulinemia at baseline was strongly associated with albuminuria at the 5-year examination.⁷⁵ A larger prospective analysis of 582 nondiabetic siblings of patients with type 2 diabetes also demonstrated that IR at baseline (assessed with glucose tolerance tests and euglycemic clamps) correlated with the development of albuminuria.⁷⁶ Interestingly, in patients with type 1 diabetes or their family members, glucose disposal rates are also reduced in the presence of albuminuria,^{77,78}

and in a small longitudinal study of patients with type 1 diabetes and normoalbuminuria, IR predicted development of albuminuria after 3 years.⁷⁹ One of the proposed mechanisms is hyperinsulinemia-mediated vasodilation of the afferent arteriole contributing to glomerular hyperperfusion and hyperfiltration (Fig. 41.4).⁸⁰

HYPERTENSION

Hypertension is common in patients with DKD. In type 1 diabetes, elevation of blood pressure is typically coincident with the appearance of albuminuria,⁸¹⁻⁸³ reflecting higher blood pressure as a consequence of DKD. Conversely, other

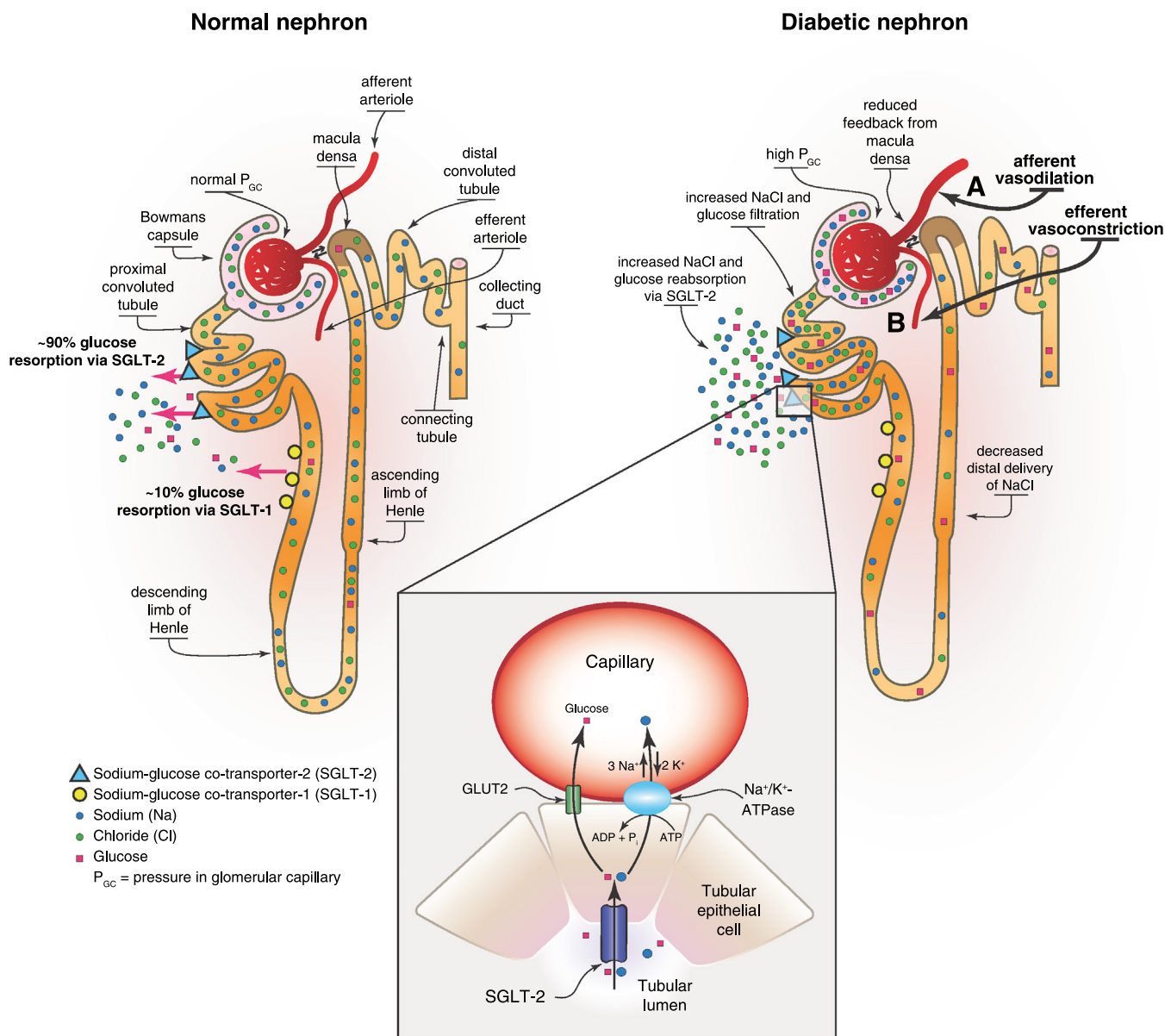


Fig. 41.4 Normal and diabetic nephron with altered hemodynamics.⁸⁰ (A) Afferent vasodilation is promoted by hyperglycemia, hyperinsulinemia, elevated levels of circulating amino acids, COX-2 prostanoids, and reduction of tubule glomerular feedback. Tubuloglomerular feedback is a kidney intrinsic autoregulatory mechanism, which helps regulate the rate of glomerular filtration rate. Because of increased reabsorption of glucose and sodium via SGLTs in diabetes, sodium chloride delivery to macula densa cells of juxtaglomerular apparatus is decreased, resulting in lower production of adenosine and consequent relative vasodilation of afferent arteriolar. (B) Efferent vasoconstriction is promoted by high local angiotensin II levels, endothelin I, reactive oxygen species, and thromboxane A₂.

studies demonstrate the predictive value of high blood pressure for DKD in type 1 diabetes. Familial aggregation of blood pressure is widely reported and may represent a predisposition to hypertension. Higher blood pressure in the nondiabetic parents of patients with type 1 diabetes and albuminuria suggests that this predisposition to hypertension is also a risk factor for DKD in type 1 diabetes.⁸⁴⁻⁸⁷ In one study, maternal blood pressure was more strongly related to albuminuria in the offspring than paternal blood pressure, suggesting a dominant effect of maternal genes or an effect of the intrauterine environment on the risk of microalbuminuria.⁸⁸ In addition, increased nocturnal blood pressure predicts the development of microalbuminuria in adolescents and young adults with type 1 diabetes.⁸⁹

In type 2 diabetes, the onset of hypertension generally precedes the first clinical evidence of DKD or even the diagnosis of diabetes. In Pima Indians, blood pressure is measured at sequential research examinations that begin long before the onset of diabetes, and higher blood pressure before the onset of type 2 diabetes was strongly associated with albuminuria after the onset of diabetes.⁹⁰ In addition, the prevalence of proteinuria was significantly higher if both parents had hypertension than if only one or neither parent had hypertension.⁹¹ Increased nocturnal blood pressure is associated with microalbuminuria in type 2 diabetes, as it is in type 1 diabetes, and a baseline systolic blood pressure >140 mm Hg has been shown to increase the risk for kidney failure and death.⁹²⁻⁹⁴ In patients with newly diagnosed type 2 diabetes, each 10 mm Hg increase in mean systolic blood pressure was associated with a 13% ($P < 0.0001$) increase in the hazard ratio for development of microalbuminuria and macroalbuminuria, eGFR <60 mL/min/1.73 m², or doubling of the plasma creatinine level after a median follow-up of 15 years. Treatment to a target blood pressure of <150/85 mm Hg compared with treatment to a target blood pressure <180/105 mm Hg resulted in a significant reduction in overall risk of microvascular complications (37%, $P = 0.009$) in the UKPDS.⁹⁵ In contrast to the legacy effects observed with intensive glycemic control, benefits of blood pressure control are transient and are lost if blood pressure rises again.

LIPIDS

Many of the abnormalities in plasma lipoproteins associated with CKD are sequelae of kidney dysfunction, but dyslipidemia may also play a role in the pathogenesis of glomerular injury.⁹⁶⁻⁹⁹ Persons with diabetes and CKD typically have significant serum hypertriglyceridemia, elevated low-density lipoprotein (LDL), and low high-density lipoprotein (HDL) cholesterol concentrations. These abnormalities are also more pronounced in persons with macroalbuminuria than in those with microalbuminuria.¹⁰⁰⁻¹⁰² Besides quantitative changes, lipid particles in persons with diabetes change qualitatively, as LDL and HDL particles tend to be smaller and denser with advancing CKD.¹⁰⁰

Among 439 patients with type 1 diabetes and proteinuria who were followed at the Joslin Clinic, elevated serum cholesterol concentration was a strong predictor of a rapid loss of kidney function.⁹⁹ Specific lipids or lipid profiles may also predict progression of DKD. In 152 patients with type 1 diabetes from the United Kingdom and Finland, elevated

serum LDL cholesterol concentration predicted progression of DKD during 8 to 9 years of follow-up, as defined by a doubling of albuminuria or a decline in creatinine clearance >3 mL/min/year.¹⁰³ Higher triglyceride content of very low-density lipoprotein (VLDL) and intermediate-density lipoprotein (IDL) particles predicted worsening of microalbuminuria, whereas smaller LDL size was associated with declining kidney function in persons with macroalbuminuria at baseline. The predictive value of these lipid profiles, however, has not been consistently observed in other studies and may conceivably reflect overall reduced access or adherence to beneficial lifestyle, dietary and medical intervention, and follow-up.^{53,104-106}

In type 2 diabetes, the UKPDS reported that higher serum LDL cholesterol concentration increased the risk for macroalbuminuria.⁶ In a post hoc analysis of 1061 persons with type 2 diabetes in the Reduction of Endpoints in Non-insulin dependent diabetes with the Angiotensin II Antagonist Losartan (RENAAL) trial, the risk of kidney failure was 32% higher for each 50 mg/dL increase in LDL cholesterol concentration and 67% higher for each 100 mg/dL increase in total cholesterol concentration. Lowering LDL cholesterol concentrations associated with the 1-year risk of kidney failure, although concurrent treatment with losartan likely contributed to this improvement.¹⁰⁷ Elevated plasma triglycerides are also associated with increased risk for both albuminuria⁶ and kidney failure in type 2 diabetes.¹⁰⁸ Triglyceride content within lipoprotein subclasses, particularly VLDL, correlates with albuminuria in youth with T1D.¹⁰⁹ In addition, low concentrations of HDL cholesterol predict increased risk of albuminuria progression in persons with type 2 diabetes and microalbuminuria.¹¹⁰ In the UKPDS, each mmol/L decline in HDL cholesterol increased the risk of doubling of serum creatinine concentration nearly threefold during a median follow-up of 15 years.⁶ Interestingly, HDL proteomic analysis in patients with kidney failure and CKD has revealed a unique proteomic profile.^{111,112}

Dyslipidemia may contribute to onset and progression of DKD through mechanisms like those responsible for atherogenesis.^{113,114} Hypercholesterolemia may impair the kidney's hemodynamic responses and tubular function by decreasing nitric oxide production and/or increasing superoxide activity in the kidney,¹¹⁵⁻¹¹⁹ with resulting antidiuretic and antinatriuretic effects.¹²⁰ Although not directly affecting GFR, these actions may play a role in the development of systemic hypertension associated with diabetes.¹²⁰ Oxidized LDL and free fatty acids can cause structural and functional damage to podocytes by inducing mitochondrial dysfunction and accumulation of reactive oxygen species,¹²¹ suggesting a causal role in the development and progression of proteinuria. Nevertheless, despite much investigation and experimental evidence supporting various mechanisms, a definitive role for dyslipidemia in the development and progression of DKD in humans remains to be established.

Besides the role of circulating lipids, the diabetic environment facilitates glomerular production of triglycerides and cholesterol, which appear to cause kidney injury both directly by accumulating in the cellular and extracellular structures and indirectly by stimulating the expression of prosclerotic, proliferative, and

proinflammatory cytokines.^{98,113,122} Accumulation of lipid droplets containing triglycerides and cholesterol has been described in kidney cells in experimental diabetes.¹²³⁻¹²⁵ These lipid droplets are associated primarily with increased expression of sterol regulatory element-binding protein (SREBP)-1c, decreased expression of peroxisome proliferator-activated receptor (PPAR)- α , and decreased expression of liver X receptor (LXR)- α , LXR- β , and ATP-binding cassette transporter-1 (*ABCA1*). Under normal conditions, *ABCA1* mediates the efflux of cholesterol to lipid-poor apolipoproteins (primarily Apo A1) to form HDLs. Down regulation of glomerular expression of *ABCA1* was observed in glomerular transcripts from persons with DKD when compared with normal controls and was correlated with lower eGFR. The downregulation of *ABCA1* was also associated with lipid droplet accumulation in podocytes,^{126,127} which in turn was found to cause cardiolipin-dependent mitochondrial dysfunction.¹²⁸ Recent studies have also challenged the relative contribution of free or esterified cholesterol to podocyte injury and proteinuria in experimental DKD.¹²⁹ A major effort was made to identify lipid content by magnetic resonance imaging (MRI), as fat content may inversely relate with eGFR and predict response to treatment strategies.¹³⁰ If and how prevention of lipid accumulation in the kidney parenchyma may affect DKD development and progression remains to be established. In this regard, the development of drugs that target Oxysterol Binding Protein Like 7 to increase cholesterol efflux, which have been shown to slow experimental CKD progression, may hold promise for clinical research.¹³¹

DIETARY PROTEIN

In experimental models, excessive protein intake causes kidney vasodilation and glomerular hyperperfusion with a resulting increase in intraglomerular pressure that leads to glomerular damage.^{132,133} Long-term high-protein intake accelerates structural and functional injury in experimental models of DKD, whereas low-protein diets offer kidney protection.¹³⁴⁻¹³⁶ Physiologic studies in humans confirm a greatly augmented glomerular hyperfiltration in response to an amino acid infusion mimicking a high-protein diet among persons with type 1 diabetes or type 2 diabetes, but only in the presence of chronic hyperglycemia (Fig. 41.4).^{137,138} Dietary protein of animal origin may also be a significant source of advanced glycation end products,¹³⁹ which could accelerate DKD progression. Meta-analyses of clinical trials that examined effects of dietary protein restriction on DKD have shown reduction in albuminuria but were inconclusive about impacts on kidney function with highly variable results between studies.¹⁴⁰⁻¹⁴²

OBESITY

The U.S. Centers for Disease Control (CDC) categorizes obesity in adults according to body mass index (BMI as kg body weight/ height m²) into class 1: BMI of 30 to <35; class 2 BMI of 35 to <40; class 3: BMI of ≥ 40 . More than one-third of adults and 17% of youth in the United States are obese. Nearly 6% of youth have class III obesity, exceeding 120% of the sex-specific 95th percentile on the CDC BMI-for-age growth charts in youth.¹⁴³⁻¹⁴⁵ Obesity is a risk factor for

CKD, independent of diabetes and hypertension. Indeed, the presence of severe obesity increases the risk of kidney failure sevenfold relative to normal weight.^{146,147} Obesity-associated hemodynamic changes in GFR and renal plasma flow appear to be responsible for the kidney damage.^{147,148} The effects of adiposity may also impact the kidneys directly through the production of various adipokines, including leptin, adiponectin, and resistin, which activate profibrotic, proliferative, and proinflammatory mechanisms.¹⁴⁹⁻¹⁵¹ A retrospective clinical and histopathologic study of 6818 native kidney biopsies found diabetes-like lesions in 45% of persons with obesity-related glomerulopathy.¹⁵² The prevalence of obesity-related glomerulopathy, defined as glomerulomegaly with or without focal segmental glomerulosclerosis, increased 10-fold between 1986 and 2000, attributed to the rising prevalence of obesity.¹⁵² Although obesity-related glomerulopathy may represent a distinct entity from DKD, obesity is largely responsible for the increase in diabetes prevalence worldwide, and similarities in underlying mechanisms of obesity-related CKD and DKD suggest that treatments targeting obesity may slow the progression of both forms of kidney disease.¹⁵³

PREGNANCY

Among women with normal kidney function, regardless of the presence or absence of diabetes, pregnancy is associated with a rise in GFR of about 50% that persists through the 37th week of gestation¹⁵⁴⁻¹⁵⁶ and is accompanied by a moderate increase in urinary protein excretion.¹⁵⁵ Among pregnant women in Austria with type 1 diabetes, those with normoalbuminuria experience a 3.8-fold increase in albuminuria, whereas those with microalbuminuria (>15 mcg/min) have a 6.7-fold increase. Nevertheless, the albuminuria levels in both groups typically normalize within 12 weeks after delivery.¹⁵⁷ Much less is known about changes in albuminuria in pregnant women with type 2 diabetes, but the prevalence of albuminuria in both types of diabetes in a Danish cohort is equivalent, and pregnancy outcomes are comparable.¹⁵⁸

Although neither pregnancy nor parity adversely affects the course of early kidney disease in women with diabetes,^{159,160} those with poor glycemic control, hypertension, or preexisting CKD are at increased risk of pregnancy complications and subsequent deterioration in kidney function.^{156,160-163} In addition, mothers with CKD are more likely to develop preeclampsia and have offspring with low birth weight.¹⁶⁴ In Finnish women with type 1 diabetes, a prior history of preeclampsia was associated with 7.7-fold higher odds of subsequent DKD than in those with normotensive pregnancies.¹⁶⁵ Preeclampsia is also associated with podocyte injury and detachment and could conceivably accelerate diabetic glomerular injury in a pregnant woman with diabetes who develops preeclampsia.¹⁶⁶⁻¹⁷⁰

INTRAUTERINE FACTORS

Fetal exposure to an abnormal intrauterine environment has lasting effects on anthropomorphic and metabolic development that lead to increased risk of CKD later in life. Certain adverse intrauterine exposures, such as a shortage of maternal fuels, various drugs, vitamin A deficiency, and

maternal diabetes, directly affect development of the fetal kidneys and reduce nephron number in the offspring, an effect that may be mediated in part via epigenetic changes from the perturbed metabolic milieu with durable effects on fetal kidney development and integrity. With the resulting reduction in filtration surface area, the kidney's adaptive capacity in response to insults, such as diabetes, is reduced, thereby enhancing the risk of CKD.¹⁷¹⁻¹⁷³ Impaired nephrogenesis may be passed on to subsequent generations through changes in epigenetic gene regulation.¹⁷⁴

Low birth weight appears to affect DKD risk in both type 1 and type 2 diabetes. In a case-control study of 184 Danish patients with type 1 diabetes, 75% of the women below the 10th percentile of birth weight had persistent macroalbuminuria (≥ 300 mg/24 hours) compared with 35% of those above the 90th percentile.¹⁷⁵ No relationship was found, however, between birth weight and urine albumin excretion in the men. In Pima Indians with type 2 diabetes, the prevalence of elevated urine albumin excretion (albumin-to-creatinine ratio ≥ 30 mg/g) was twice as high in both men and women of low birth weight than in those of normal birth weight and was three times as high in those of high birth weight.¹⁷⁶ Two-thirds of those in the high birth weight category were the offspring of diabetic mothers.¹⁶⁶ Irrespective of birth weight, preterm birth also appears to impose an additive risk for progression of kidney disease and such information should be collected as part of a patient's health history.¹⁷⁷

Fuel-mediated alterations in maternal metabolism in pregnancies complicated by diabetes may preferentially harm poorly replicating, terminally differentiated cells, such as those found in the nephron.¹⁷⁸ A proposed mechanism for this differential apoptosis during nephrogenesis is activation of the nuclear factor NF- κ B signaling pathway.¹⁷⁹ The adverse effect of hyperglycemia on nephrogenesis was demonstrated experimentally by exposing pregnant rats to diabetes and counting nephrons in the offspring. The number of nephrons was reduced by up to 35% in those exposed to diabetes, and even minor elevations of glucose concentration were associated with impaired metanephros development.¹⁸⁰ In adult offspring of women with type 1 diabetes who were exposed to maternal diabetes in utero, kidney functional reserve was significantly reduced relative to those who were not exposed, likely reflecting reduced nephron mass because of this exposure.¹⁸¹ In Pima Indians with type 2 diabetes, the odds of elevated urine albumin excretion were nearly fourfold higher in those exposed to diabetes in utero than in those who were not exposed.¹⁸² Moreover, exposure to diabetes in utero among Pima Indian offspring increased nearly fourfold over a 30-year period, paralleled by a doubling in the prevalence of childhood-onset diabetes attributable to this exposure.¹⁸³ These data point to in utero diabetes exposure as a contributor to the rising prevalence of childhood and youth-onset type 2 diabetes. Intrauterine exposure to diabetes was also associated with a fourfold increase in the age-sex-adjusted incidence of kidney failure in young adults with type 2 diabetes, mediated largely by the younger age at onset of diabetes.¹⁸⁴ These observations have potential relevance in the developing world and in disadvantaged groups in developed countries, where a higher proportion of type 2 diabetes develops during the childbearing years.¹⁸⁵ The

greater frequency of diabetes during the childbearing years in these populations means that the offspring are more likely to be exposed to a diabetic intrauterine environment and to suffer consequences of that exposure.

SMOKING

The role of smoking in promoting DKD is unclear, but substantial evidence supports an effect on albuminuria,¹⁸⁶⁻¹⁹¹ although a relationship with progressive loss of GFR is less clear.^{58,192} For instance, smoking increased the risk of albuminuria after 5 years of follow-up in a population-based cohort of 3667 Swedish persons with type 2 diabetes who had no kidney disease at baseline, but it was not associated with GFR decline to <60 mL/min/1.73 m² during follow-up.¹⁸⁸ In the Finnish Diabetic Nephropathy Study, the 12-year cumulative risk of macroalbuminuria and kidney failure in 3613 patients with type 1 diabetes were 14.4% and 10.3% for current smokers, 6.1% and 10.0% for ex-smokers, and 4.7% and 5.6% for nonsmokers.¹⁹³

Nicotine induces podocyte apoptosis through increasing oxidative stress, promotes mesangial cell proliferation and extracellular matrix production, and increases transforming growth factor (TGF)- β and fibronectin expression in experimental models.¹⁹⁴⁻¹⁹⁶ Given that persons with diabetes already have widespread vascular damage because of their diabetes,¹⁹⁷ smoking only accelerates this damage. The long-term risks of electronic cigarettes are also of concern, as possible nephrotoxicity of e-cigarette refill liquid was observed in rat kidneys.¹⁹⁸

PERIODONTAL DISEASE

Periodontal disease is an inflammatory condition, which often occurs in the absence of diabetes, but is also a frequent complication of diabetes,¹⁹⁹ contributing to poor glycemic control, low-grade chronic systemic inflammation, and increased risk of macrovascular and microvascular complications.^{199,200} Among patients with CKD, periodontal disease substantially increases the risk of premature mortality.²⁰¹ In a Swedish case-control study of 78 patients with type 1 or type 2 diabetes, those with severe periodontitis, based on alveolar bone loss, had a higher frequency of dipstick-positive proteinuria and cardiovascular complications after 6 years than those with mild periodontal disease.²⁰² Severity of periodontitis by alveolar bone loss and being edentulous predicted both albuminuria and kidney failure in a dose-dependent manner among 529 Pima Indian adults with type 2 diabetes who were followed for a median of 9 years.²⁰³ A study investigating the relationships among diabetes, periodontal disease, and CKD in 11,211 adults from the NHANES 1988–1994 population suggested a bidirectional relationship between CKD and periodontal disease. Periodontitis increased the risk of CKD, while CKD also increased risk of periodontitis.²⁰⁴ Although the mechanisms linking periodontal disease to kidney damage remain to be established, serum lipopolysaccharide activity, which is induced by bacterial infections, is associated with the progression of DKD in Finnish patients with type 1 diabetes.²⁰⁵

Control of periodontal infection in diabetic adults improves A1c level¹⁹⁹ and reduces the concentration

of various markers of inflammation, coagulation, and adhesion.²⁰⁶ Whether such control also reduces the onset or progression of DKD is not known.

CARDIAC AUTONOMIC NEUROPATHY

Abnormalities in heart rate control and vascular dynamics associated with diabetes are referred to as cardiac autonomic neuropathy and are the consequence of damage to and loss of the small unmyelinated nerve fibers that innervate the heart and blood vessels. Advanced cardiac autonomic neuropathy may present as resting tachycardia and orthostatic hypotension. The presence of cardiac autonomic neuropathy is a risk factor for DKD in several longitudinal studies of type 1 and type 2 diabetes. Among 35 Swedish patients with type 1 diabetes, those with cardiac autonomic neuropathy had a significant decline in GFR measured by ⁵¹Cr-EDTA clearance over a 10-year period, whereas those without cardiac autonomic neuropathy experienced almost no change.²⁰⁷ Cardiac autonomic neuropathy was strongly associated with both early GFR loss and progression to CKD, defined by estimated GFR <60 L/min/1.73 m², in a subset of the First Joslin Kidney Study, which included 204 normoalbuminuric patients with type 1 diabetes and 166 with microalbuminuria who were followed for a median of 14 years.²⁰⁸ Similar associations were reported in a multiethnic cohort of 204 adults with type 2 diabetes from the United Kingdom, in whom estimated GFR declined more rapidly over 2.5 years in those with cardiac autonomic neuropathy (9.0% decline) than in those without (3.3% decline)²⁰⁹ and in a cohort of 1117 Korean patients with type 2 diabetes and estimated GFR ≥60 mL/min/1.73 m² at baseline, in whom the presence of cardiac autonomic neuropathy increased the risk of developing CKD by over 2.6-fold during nearly 10 years of follow-up.²¹⁰ A cross-sectional study in 63 Pima Indians who underwent research kidney biopsies demonstrated an association between cardiac autonomic neuropathy and the structural lesions of early progressive DKD.²¹¹

FAMILIAL AND GENETIC FACTORS

Familial clustering of DKD, as well as population ancestry and ethnic differences in disease susceptibility, suggest a genetic predisposition with a 2.1- to 2.3-fold risk for DKD in type 1 diabetes siblings with DKD.²¹²⁻²¹⁴ Narrow-sense DKD heritability, defined as the proportion of phenotypic variance explained by additive effects of genotyped SNPs, ranged between 24% and 59% in unrelated individuals with type 1 diabetes from the FinnDianne Cohort.^{215,216} Similar analyses of individuals with type 2 diabetes suggested only 8% to 25% heritability, which may potentially reflect that more heterogeneous mechanisms contribute to DKD in these patients and possibly a more important contribution of environmental factors.²¹⁷

Several candidate genes have been identified that may be related to DKD. One of the most intensively studied candidate genes is the insertion/deletion (I/D) polymorphism of the ACE gene (*ACE/ID*). Among 168 Japanese patients with type 2 diabetes who were followed for 10 years, analysis of the time course of the three ACE genotypes indicated that patients with the DD genotype were more likely to progress

to kidney failure during follow-up than those with the other genotypes, and they had higher mortality once dialysis was initiated.²¹⁸ The deleterious effect of the D allele on kidney function was subsequently confirmed in patients from other patient groups with type 2 diabetes²¹⁹⁻²²¹ and in type 1 diabetes.^{222,223} Presence of the D allele was also associated with more severe structural lesions in type 2 diabetes²²⁴ and with greater progression of structural lesions in type 1 diabetes.²²⁵ In the RENAAL trial, the presence of the D allele increased the likelihood of reaching the primary endpoint of doubling of baseline serum creatinine concentration, kidney failure, or death in the placebo group, but treatment with losartan mitigated that risk, suggesting that losartan had its greatest effect in patients with the D allele.²²⁶ Another candidate gene related to the renin-angiotensin system was identified by the Bergamo Nephrologic Diabetes Complications Trial (BENEDICT), which found that the Pro618Ala polymorphism of the *ADAMTS13* gene was associated with a higher risk of CKD progression in patients with type 2 diabetes, but patients with this polymorphism also had a better response to ACE inhibitors.²²⁷

Kidney failure is considered by some investigators to be the optimal phenotype of DKD in genetic association studies,^{228,229} so the present discussion focuses on this phenotype. Genome-wide association studies (GWASs) aim to identify associations between a genotype and a phenotype. While there is evidence of heritability of DKD, only a few loci associated with DKD, albuminuria, or eGFR in subjects with diabetes have been identified. In a GWAS study of pooled genomic data, the plasmacytoma variant 1 (*PVT1*) gene was identified as a potential susceptibility locus for kidney failure in the Pima Indians²³⁰ and this association was subsequently confirmed in the Genetics of Kidneys in Diabetes (GoKinD) study among persons of European descent with type 1 diabetes.²³¹ A GWAS study also involving the GoKinD cohort and confirmed in the DCCT/EDIC cohort, found SNPs in the FERM domain-containing protein 3 (*FRMD3*) gene and near the cysteinyl-tRNA synthetase (*CARS*) gene associated with DKD, defined by overt proteinuria or kidney failure.²³² A GWAS meta-analysis of 6691 patients with type 1 diabetes and DKD performed by the GENIE consortium (Genetics of Nephropathy: an International Effort) identified two SNPs in the *AFF3* and an SNP between the *RGMA* and *MCTP2* genes associated with kidney failure. Thereby, *AFF3* seems to contribute to renal tubule fibrosis via the TGF-β1 pathway. The strongest association with DKD as a primary phenotype was seen for an intronic SNP in the *ERBB4* gene (rs7588550).²¹⁶ A subsequent analysis in the Joslin Study of Genetics of Nephropathy in Type 2 Diabetes found that susceptibility loci near *CARS* were common to both types of diabetes.²³³ The Family Investigation of Nephropathy and Diabetes (FIND) conducted a genome-wide association study in 6197 European American, African American, American Indian, and Hispanic American families in whom type 2 diabetes was the predominant cause of kidney disease.²³⁴ Index cases all had diabetes for more than 5 years or diabetic retinopathy, with a UCAR >1 g/g or kidney failure; unrelated controls had diabetes for at least 9 years and normal UCARs. A replication cohort included 7539 additional European American, African American, and American Indian cases and nonnephropathy controls. The

FIND investigators found and replicated a DKD-associated genetic locus on chromosome 6q25.2 (rs955333) between the *SCAF8* and *CNKSR* genes across persons of different ancestries. Findings were supported by significantly different gene expression patterns in this region from kidney tissue in persons with DKD versus those without. Extensive work in this field has identified several genomic areas of interest, but results thus far only support a role for multiple susceptibility genes, each with weak effects.

In a meta-analysis of 19,406 subjects with type 1 diabetes by the Diabetic Nephropathy Collaborative Research Initiative (DNCRI) consortium, only four genome-wide loci were significantly associated with DKD (i.e., *COL4A3*, *BMP7*, *COLEC11*, and *DDR1*). Interestingly, the *COL4A3* common missense mutation rs55703767 (Asp326Tyr) was associated with a 21% lower risk for DKD.²³⁵

Phenome-Wide Association Studies (PheWAS) is a novel approach where multiple phenotypes are analyzed in comparison with a single genetic variant. Unlike a GWAS, in which a genotype is associated with a phenotype, this approach tries to associate phenotypes with genotypes. PheWAS was originally designed to use phenotype records obtained from electronic health records as a tool for genomic investigations.²³⁵ While PheWAS has been used to confirm the associations, for example, of XOR variants in patients at high risk of developing DKD,²³⁶ more research is needed to unambiguously establish phenotype-genotype associations in DKD. Nevertheless, this approach is promising and may also provide a tool to determine long-term consequences of drug target manipulation.

The Type 2 Diabetes Knowledge Portal (T2DKP) is a public resource of genetic datasets and genomic annotations focused on type 2 diabetes-related traits, which aims to provide supporting evidence for the role of certain genes in disease progression, thereby increasing their viability as therapeutic targets.²³⁷

EPIGENETIC FACTORS

In addition to genetic factors, multifaceted crosstalk between genes and environmental factors can induce tissue-specific epigenetic changes and mechanisms. These changes include DNA cytosine methylation, histone posttranslational modifications in chromatin, and noncoding (nc) ribonucleic acids (RNAs), all of which can modulate diabetes complications through alterations in gene expression.²³⁸ Epigenome-wide association studies (EWAS) revealed differentially methylated regions localized in a large number of genes in patients with type 1 and type 2 diabetes that are associated with DKD. Through epigenetic mechanisms, cells acquire metabolic memory of prior hyperglycemic exposure that may modulate development and progression of DKD.^{239,240} Histone post translational modification studies, in particular at key metabolic genes suggest a crucial role for this process in the pathogenesis of DKD. Hyperglycemia-induced epigenetic changes activate transcription factors involved in the expression of genes mediating the pathogenesis of DKD.²³⁸ In addition, cytosine methylation changes in genes related to kidney fibrosis in human kidney tissue correlate with downstream transcript levels and provide further evidence that epigenetic dysregulation plays a role in the development of CKD.²⁴¹

Similarly, posttranslational modifications of ncRNAs, certain long noncoding RNAs, and miRNAs also have regulatory roles in DKD including promoting/modulating fibrotic gene expression in kidney cells by targeting transcription repressors.²³⁹ Although the mechanisms of such cellular memory are not entirely known, its presence is supported by the regression of morphologic lesions in diabetic kidneys after a prolonged period of normoglycemia following pancreas transplantation.²⁴² Similarly, the long-lasting effects of previous strict glycemic control observed in persons with type 1 diabetes in the DCCT²⁴³ or with type 2 diabetes in the UKPDS²⁴⁴ may be due to metabolic memory.

ALTERATIONS IN GUT MICROBIOME

Recent studies report the possible role of the gut microbiota in development and progression of DKD. Interactions between gut microbiota and the kidney, constituting the so-called “gut–kidney axis,” is maintained via complex cellular and molecular signaling. Metabolomic studies have demonstrated significant differences in gut microbiota diversity among the patients with diabetes and DKD compared with healthy controls.^{245,246} Furthermore, in patients with biopsy-confirmed DKD, a recent study reported the presence of microorganisms positively correlated with glomerulosclerosis, glomerular basement membrane thickening, and albuminuria.²⁴⁷ The exact mechanism by which gut microbiota exert the influence on DKD risk remains incompletely understood. One of the proposed mechanisms is microbiome disruption associated with a deficiency of short-chain fatty acids, which play a vital role in reducing inflammation.²⁴⁸

Clinical Relevance: Risk Factors

Multiple risk factors have been associated with an increased risk for progression to DKD. Key risk factors include, but are not limited to, suboptimal glycemic control, hypertension, dyslipidemia, obesity, smoking, intrauterine factors, epigenetic and genetic factors, and dietary patterns. Optimizing management of modifiable risk factors (e.g., glycemia, blood pressure, and lipids) represent important approaches to prevention of CKD.

BIOMARKERS OF DIABETIC KIDNEY DISEASE

Albuminuria is the best available risk marker for DKD, but it has limitations. Not only does albuminuria return to normal spontaneously or in response to therapy in many patients with diabetes,^{51,249-254} but the absence of albuminuria does not preclude the presence of DKD.^{5,6,38,52} Therefore other biomarkers that provide additional prognostic or mechanistic information for progressive DKD may help to enrich clinical trial populations by selectively enrolling those at highest risk or with specific pathway activation targeted for therapeutic intervention.^{255,256} A comprehensive list of potential biomarkers is shown in Table 41.2.²⁵⁷ Biomarkers fall into several classes that reflect underlying pathogenic

Table 41.2 Biomarkers of Diabetic Kidney Disease

Class	Biomarkers	Method of detection
Currently used	GFR	Renal clearance/ estimating equation
	Albuminuria	Urine
	Creatinine	Serum
	Cystatin C	Serum
Oxidative stress	BUN	Serum
	Pentosidine	Serum/urine
	8-OHdG	Urine
	Uric acid	Serum
	AGEs/OPs	Serum
	IPP2K	Urine
Fibrosis	Adiponectin	Serum
	TGF- β_1	Serum/urine
	CTGF	Serum/urine
	VEGF	Serum/urine
Glomerular damage	Transferrin	Urine
	Type IV collagen	Urine
Tubular damage	Cystatin C	Urine
	L-FABP	Urine
	NGAL	Urine
	KIM-1	Serum/urine
	ACE2	Serum/urine
	Angiotensinogen	Urine
	NAG	Urine
	$\alpha 1$ -microglobulin	Urine
	FGF23	Serum
	EGF	Urine
Inflammation	TNF- α ; TNFR 1/2	Serum/urine
	Osteoprotegerin	Plasma
	MCP-1	Urine
	IL-1, IL-6, IL-8, IL-18	Serum/urine
	WBC counts/ fractions	Blood
	Bradykinin and related peptides	Plasma
	hs-CRP	Serum
	Osteopontin	Serum
	β -trace protein	Serum
	β -2 microglobulin	Serum
Mitochondrial function	Various metabolites	urine

NOTE: This table is not intended to be an exhaustive list of potential biomarkers but is intended to illustrate the diversity of biomarkers and mechanisms.

8-OHdG, 8-hydroxy-2'-deoxyguanosine; ACE2, Angiotensin-converting enzyme-2; AGE, advanced glycation end product; BUN, blood urea nitrogen; CTGF, connective tissue growth factor; EGF, epidermal growth factor; FGF23, fibroblast growth factor 23; GFR, glomerular filtration rate; hs-CRP, high-sensitivity C-reactive protein; IL, interleukin; IPP2K, inositol pentakisphosphate 2-kinase; KIM-1, kidney injury molecule 1; L-FABP, liver fatty acid-binding protein; MCP-1, monocyte chemoattractant protein-1; NAG, N-acetyl- β -D-glucosaminidase; NGAL, neutrophil gelatinase-associated lipocalin; OP, oxidative product; TGF- β_1 , transforming growth factor- β_1 ; TNF- α , tumor necrosis factor- α ; TNFR 1/2, tumor necrosis factor receptor 1 and 2; VEGF, vascular endothelial growth factor; WBC, white blood cell.

Adapted and updated from Tables 1 and 2 from Campion CG, Sanchez-Ferraz O, Batchu SN. *Can J Kidney Health Dis.* 2017;4:1–18 with permission.²⁵⁷

mechanisms, particularly markers of tubular damage, inflammation, and oxidative stress, as well as panels that combine markers.

Much attention has focused on urine concentrations of molecules predominantly expressed by renal tubular cells as DKD biomarkers. These molecules associate strongly with acute kidney injury, which may accelerate the progression of DKD.²⁵⁸⁻²⁶⁰ Tubular markers such as kidney injury molecule (KIM-1), liver-fatty acid binding protein (L-FABP), N-acetyl- β -D-glucosaminidase (NAG), and neutrophil gelatinase-associated lipocalin (NGAL) have been evaluated in relation to DKD, cardiovascular disease, and mortality.²⁶¹⁻²⁷⁴ Inconsistencies in these results may be due to differences in study design, use of surrogate or composite outcomes, or incomplete covariate adjustment in risk models. A recent study also found that the plasma concentration of KIM-1 was a stronger predictor of early progressive kidney function loss in type 1 diabetes than the urine KIM-1 concentration, suggesting that its source may reflect distinct aspects of proximal tubular damage.²⁷⁵ Although these studies suggest that damage to the proximal tubules is important in DKD, these markers do not meaningfully enhance risk prediction over established risk markers.

Another tubular marker that has attracted recent attention is epidermal growth factor (EGF), a growth-promoting peptide synthesized in kidney tubular cells and found in the urine. Lower urine excretion of EGF was found in persons with type 1 diabetes and elevated urine albumin excretion than in nondiabetic controls. Among persons with diabetes and GFR >90 mL/min, urine EGF correlated directly with GFR and inversely with urine albumin excretion.²⁷⁶ In addition, lower urine EGF-to-creatinine ratios were associated with incident-impaired eGFR, >5% loss of eGFR per year, or both in persons with type 2 diabetes who were normoalbuminuric with preserved kidney function at baseline.²⁷⁷ Transcriptomic data obtained from kidney tissue in three CKD cohorts, which included persons with diabetes, were used to identify potential CKD biomarkers. The top candidate identified by this approach was EGF, and the lower urine concentrations of EGF in these patients improved prediction of progressive CKD beyond established risk factors.²⁷⁸ Such a transcriptomic-driven sequential strategy used for biomarker discovery and identification will be used for ongoing biomarker research.

Inflammatory processes also play a key role in DKD, and several promising biomarkers related to inflammation have been identified. Tumor necrosis factor- α (TNF- α) was implicated in the pathogenesis of DKD in experimental models in the early 1990s,²⁷⁹ but its association with DKD in humans has not been convincingly established. On the other hand, circulating levels of tumor necrosis factor receptors (TNFR) 1 and 2 have emerged as robust and independent predictors of DKD progression. In the Joslin Clinic diabetes cohorts, elevated concentrations of circulating TNFR 1 and 2 were strongly associated with impaired eGFR (<60 mL/min/1.73 m²) in patients with type 1 diabetes²⁸⁰ and kidney failure in patients with type 2 diabetes,²⁸¹ after controlling for established risk factors. Elevations of TNFRs were subsequently found to predict macroalbuminuria in patients with type 1 diabetes in the DCCT/EDIC cohort,²⁸² kidney failure in the FinnDiane type 1 diabetes cohort,²⁸³ loss of kidney function in the SURDIAGENE type 2 diabetes cohort,²⁸⁴

and kidney failure in Pima Indians with type 2 diabetes.²⁸⁵ TNFRs also predicted cardiovascular events and total mortality in Swedish patients with type 2 diabetes from the CARDIPP study.²⁸⁶ Research kidney biopsies performed in Pima Indians with type 2 diabetes demonstrated strong associations between higher concentrations of TNFR 1 and 2 and various morphometric lesions including mesangial expansion, loss of endothelial cell fenestration and total filtration surface per glomerulus, increased width of the glomerular basement membrane and podocyte foot processes, and increased global glomerular sclerosis, suggesting that they may be involved in the pathogenesis of early glomerular lesions in DKD.²⁸⁷ Proteomic profiling of 194 serum proteins in patients of three cohorts with type 1 and type 2 diabetes identified a kidney risk inflammatory signature (KRIS) of 17 proteins. Among them are six TNFR Superfamily members including TNFR1 and TNFR2 but also other inflammatory proteins such as CCL14, CCL15, and CSF1. Serum KRIS proteins were found significantly higher in diabetic patients who progressed to kidney failure, and increased serum KRIS protein levels were associated with an increased 10-year risk of kidney failure in the Joslin cohort of patients with type 1 diabetes and in Joslin and Pima cohorts of patients with type 2 diabetes.²⁸⁸

Monocyte chemoattractant protein-1 (MCP-1) is a cytokine secreted by mononuclear leukocytes, cortical tubular epithelial cells, and podocytes that is implicated in inflammatory processes in the kidneys that lead to fibrosis.²⁸⁹ Higher urine MCP-1 concentrations are associated with increased risk of kidney function decline, doubling of serum creatinine concentration, and progression to dialysis or death in patients with DKD and either type 1 or type 2 diabetes.²⁹⁰⁻²⁹² MCP-1 is also one of several urine cytokines that predict early decline in GFR in patients with type 1 diabetes and microalbuminuria.²⁹³ Urine MCP-1 correlates with early cortical interstitial expansion in normotensive normoalbuminuric individuals with type 1 diabetes, suggesting that it may be involved in the pathogenesis of early interstitial changes in DKD.²⁹⁴

Risk stratification may be improved by assessing markers and marker species (proteins, lipids, metabolites) from several different pathways simultaneously.^{256,295-299} A study involving a subset of participants with early DKD from the Action to Control Cardiovascular Risk in Diabetes (ACCORD) trial and with advanced DKD from the Veterans Administration NEPHropathy iN Diabetes (VA-NEPHRON-D) study illustrated that a combination of inflammatory (plasma TNFR 1 and 2) and tubular (plasma KIM-1) biomarkers were independently associated with a higher risk of eGFR decline in type 2 diabetes. Importantly, together these markers significantly improved risk prediction over established markers.³⁰⁰ Similarly, the Genetics of Diabetes Audit and Research Tayside Study (GO-DARTS) explored a set of 207 serum proteins and metabolites in a nested case-control study of a subset of patients with type 2 diabetes and stage 3 CKD to identify biomarkers that predicted rapid kidney function decline, defined by >40% decline in eGFR within 3.5 years. Fourteen biomarkers from across several pathogenic pathways differentiated the cases with rapid kidney function decline from the controls, who had no fall in estimated GFR, and these markers improved

prediction of rapid progression over established risk factors.³⁰¹

Plasma apolipoprotein M (pAPOM) has also emerged as a potential biomarker for DKD. pAPOM levels were significantly lower in patients with type 2 diabetes when compared with controls.^{302,303} pAPOM levels also correlated with the clinical stage of DKD in 104 patients with type 2 diabetes and were associated with disease progression in a 5-year observation period.³⁰⁴ Meanwhile, in 386 patients with type 1 diabetes of DCCT/EDIC cohort, increased pAPOM levels associated with progression to macroalbuminuria and CKD.³⁰⁵ These seemingly opposing observations may indicate that different pathways regulate pAPOM levels in patients with type 1 diabetes and type 2 diabetes. Of note, reduced pAPOM levels were also found in patients with maturity-onset diabetes of the young (HNF1A-MODY) when compared with controls or subjects with type 1 diabetes but similar to patients with type 2 diabetes.³⁰⁶ Reduced pAPOM may help to distinguish HNF1A-MODY patients from patients with type 1 diabetes.

Novel omics technologies such as proteomics, lipidomics, and metabolomics and the implementation of machine learning have revolutionized biomarker research. These technologies offer the opportunity to analyze large sets of markers and study their association with clinical variables such as albuminuria or eGFR decline. In the Outcome Reduction with Initial Glargine Intervention (ORIGIN) trial, a large protein panel of 239 biochemical markers was used to identify prognostic markers of eGFR decline in 7482 patients with dysglycemia. Findings of this study indicated that the addition of 15 biomarkers to routinely measure clinical markers significantly improves the prediction of eGFR.³⁰⁷ The analysis of plasma samples from 258 patients selected from a cohort of patients with type 2 diabetes and incident CKD and stable eGFR and of 223 patients with rapid eGFR decline (median eGFR decline -6.75 mL/min/year, median follow-up 35 months) of the PROVALID study (PROspective cohort study in patients with type 2 diabetes for VALIDation of biomarkers) by proteomics, lipidomics, and metabolomics identified KIM-1 and NTProBNP as best predictors of eGFR decline. Interestingly, this study also revealed that the addition of lipidomic and metabolomic analyses did not further improve the predictive value of these markers, nor did it lead to the identification of additional novel markers of eGFR decline.³⁰⁸ Using plasma and urine samples of 135 patients with DKD in nontargeted metabolomics combined with comprehensive machine learning, 6 metabolites including urinary threonic acid and plasma CE-C-0218 were found to predict rapid eGFR decline with superior accuracy than albuminuria. Nevertheless, the small sample size ($n = 14$) of patients with rapid eGFR decline in this study warrants further investigations.³⁰⁹ Combined proteomic and metabolomic analyses of sera from 1513 patients with type 2 diabetes and patients with early- and advanced-stage DKD identified α_2 -macroglobulin, cathepsin D, and CD324 as robust biomarkers of DKD progression, as well as the metabolite glycerol-3-galactoside as an independent metabolomic biomarker of DKD progression.³¹⁰ Using a combination of metabolomics, lipidomics, genome-wide SNP genotyping, and machine learning to analyze samples of 618 patients with diabetes, nondiabetic CKD, and DKD, a biomarker signature consisting of phosphatidylcholine,

lysophosphatidylcholine, resolvin D1, and pseudouridine was found to associate with DKD. To date, GWASs have identified more than 30 genomic DNA polymorphisms to be associated with DKD.³¹¹ Nevertheless, while SNPs in several genes such as *FRMD3*,³¹² *COL4A3* (rs55703767),³¹² *CNTNAP2* (rs1989248),²¹⁶ *REF AFF3* (rs7583877),³¹³ and *ERBB4* (rs7588550) associate with DKD, a combination of these SNPs with traditional risk factors was not found to better discriminate between DKD and nondiabetic CKD.³¹⁴

Mitochondrial dysfunction has been implicated in the pathogenesis of DKD, and several mitochondrial pathways are modulated in diabetes. Excess generation of reactive oxygen species (ROS), followed by the oxidative damage of mitochondrial proteins and mitochondrial DNA (mtDNA), is one of the main contributors to mitochondrial dysfunction in the diabetic kidney. Studies have shown that the level of mtDNA is increased in the peripheral blood, urine, and kidney biopsies of patients with DKD.³¹⁵

Various microRNAs,^{316,317} long noncoding RNAs,²³⁰ and urine exosomes³¹⁸ may be informative biomarkers for DKD. Recent advances in imaging technology may also make it possible to improve risk stratification of early DKD using MRI-based technologies³¹⁹⁻³²¹ and optical coherence tomography, which predicts 4-year incident diabetic neuropathy³²² and is being evaluated for use in DKD.

NATURAL HISTORY AND CLINICAL COURSE

The traditional model for the natural history of DKD centers around development of glomerular hyperfiltration that progresses to increasing albuminuria, followed by declining GFR.³²³ Supraphysiologic elevation in GFR is observed early in the natural history of patients with both type 1 and type 2 diabetes.³²⁴ Although this change is not present in all patients, the presence of glomerular hyperfiltration in early diabetes is often associated with the development of more severe kidney damage at the later stages of DKD.³²⁴⁻³²⁶ A meta-analysis of 780 patients with type 1 diabetes from 10 cohort studies found the odds of developing albuminuria in those with glomerular hyperfiltration at baseline was 2.7 times that of persons without it at baseline,³²⁵ and a study of 600 patients with type 2 diabetes reported that those with persistent glomerular hyperfiltration during a 6-month period from baseline examination were more than twice as likely to develop microalbuminuria during 4 years of follow-up than those without hyperfiltration.³²⁷

Nonetheless, a sizable proportion of persons with diabetes do not follow the classical paradigm. In these individuals, DKD can present with low eGFR and without albuminuria.^{38,328,329} For instance, over a median follow-up of 15 years of newly diagnosed type 2 diabetes in the UKDPS, 40% developed albuminuria and 30% progressed to an eGFR <60 mL/min/1.73 m² or experienced a doubling in serum creatinine. Of participants experiencing a decline in kidney function, 60% did not have preceding albuminuria.⁶ In DCCT/EDIC, out of the 11% of participants with type 1 diabetes who developed an eGFR <60 mL/min/1.73 m² during a mean follow-up of 19 years, 40% did not develop macroalbuminuria before they developed reduced GFR and 24% never exhibited albuminuria.³⁸ Further support for this observation came from the Developing Education on Microalbuminuria for

Awareness of Renal and Cardiovascular Risk in Diabetes (DEMAND) study, with approximately 21%, 31%, and 35% of individuals with normoalbuminuria, microalbuminuria, and macroalbuminuria, respectively, who developed a creatinine clearance <60 mL/min/1.73 m².⁵ The clinical presentation of DKD has shifted such that the prevalence of low eGFR in the absence of albuminuria is observed more commonly. Indeed, a study comparing DKD presentation during the time periods of 1988–1994 and 2009–2014 reported that the prevalence of albuminuria decreased from 21% to 16%, while the prevalence of low eGFR (<60 mL/min/1.73 m²) and severely reduced GFR (<30 mL/min/1.73 m²) increased from 9% to 14%, and 1% to 3%, respectively.³³⁰ The classical paradigm of DKD is further challenged by findings that albuminuria is dynamic and fluctuating rather than linearly progressive. For example, during nearly 8 years of follow-up in the Multifactorial Intervention for Patients with Type 2 Diabetes Study (Steno-2), 31% of participants with microalbuminuria progressed to macroalbuminuria, 38% remained microalbuminuric, and 31% regressed to normoalbuminuria.²⁵³ Studies of persons with type 1 diabetes and microalbuminuria from the Joslin Clinic showed that only 19% developed macroalbuminuria while 59% regressed to normal albuminuria after 6 years of follow-up.²⁵⁰ Similarly, among participants from DCCT/EDIC who developed microalbuminuria that remained for two consecutive visits, 28% subsequently progressed to macroalbuminuria and 40% regressed to normoalbuminuria after 10 years.²⁴⁹ Another study demonstrated remission of microalbuminuria to normoalbuminuria in 24% of cases and regression of macroalbuminuria to microalbuminuria in 30% of cases.³³¹

Only limited data on the natural history of DKD by diabetes type are available. In general, albuminuria rates appear to be similar in patients with type 1 and type 2 diabetes, with some studies showing that low eGFR is more common in type 2 diabetes.³³² However, an analysis of the Diabetes Swedish Full Population Study (DAISY) and Diabetes Norwegian Full Population Study (DAPH-NE) questions these findings. This analysis included a total of 59,331 patients with type 1 diabetes and 484,241 patients with type 2 diabetes. After a mean period of 2.6 years, adult patients with type 1 diabetes had the greater risk of kidney disease with a 1.4- to 3-fold higher risk at all ages compared with those with type 2 diabetes.³³³ In terms of progression to kidney failure, rates of eGFR decline can be highly variable between patients. Indeed, data are also mixed regarding whether the incidence of kidney failure differs by type of diabetes.^{334,335} The data from Australia's National Diabetes Services Scheme showed that the age-standardized annual incidence of kidney failure increased for people with type 2 diabetes and remained stable for patients with type 1 diabetes from 2002 to 2013.³³⁵

The presence of albuminuria is strongly associated with hypertension. In type 2 diabetes, 40% to 83% of patients with microalbuminuria and 78% to 96% of patients with macroalbuminuria are hypertensive.³³⁶ In type 1 diabetes, the prevalence of hypertension sharply increases from 30% in those without kidney disease to a prevalence that parallels that seen in type 2 diabetes once DKD develops.^{337,338} The overall risk of cardiovascular disease (heart failure, myocardial infarction, and stroke) increases from age 40 years in both types of diabetes, with a slightly higher prevalence in

type 1 diabetes when compared with type 2 diabetes (heart failure event rates in patients with type 1 diabetes aged 65–79 years were higher [HR 1.31–1.39]; rates of stroke in patients with type 1 diabetes aged 40–54 years were HR 1.35–1.72; total cardiovascular disease events at ages 55–79 years: HR 1.14–1.42 compared with similar-aged patients with type 2 diabetes) (Fig. 41.5).³³³ DKD is a strong predictor of mortality, predominantly from cardiovascular disease. In the absence of DKD, patients with type 1 diabetes have a life expectancy comparable with that of the general population.^{339,340} The FinnDiane Study and Pittsburgh Epidemiology of Diabetes Complications Study extended these findings to show a graded association between the severity of kidney disease and mortality. In addition, they found that excess mortality in type 1 diabetes, relative to the background population, was almost entirely observed in patients with albuminuria.^{340,341} Similar findings were reported in type 2 diabetes with higher albuminuria and lower eGFR independently and additively associated with an increased risk for cardiovascular events, mortality, and all-cause death.^{13,342,343} Using nationally representative data from NHANES III, 10-year mortality in patients with type 2 diabetes in the United States was examined by level of kidney impairment and compared with mortality in persons without diabetes and kidney disease. Relative to this reference group, most of the excess mortality associated with type 2 diabetes was found in patients with

DKD, with the greatest excess noted in those who had both albuminuria and reduced eGFR.¹³ While both types of diabetes increase risk of cardiovascular disease compared with the general population, analyses of Sweden and Norwegian National Data Registries showed that between 2013 and 2016 cardiovascular mortality rates were higher in type 1 diabetes than type 2 diabetes patients at ages 55 to 64 years and 70 to 79 years (HR 1.37–1.71) (Fig. 41.5).^{333,344}

CKD complications occur earlier and with greater frequency and severity in diabetes. Due to a prominent tubulointerstitial component of DKD, damage to erythropoietin-producing peritubular interstitial cells occurs with consequent erythropoietin deficiency. Therefore patients with DKD are nearly twice as likely to have anemia compared with patients with nondiabetic CKD at comparably reduced levels of eGFR.³⁴⁵ IR in patients with diabetes contributes to lower parathyroid hormone levels with an increased predisposition to adynamic bone disease (Fig. 41.6).^{345,346}

PATHOLOGY: STRUCTURAL CHANGES OF THE KIDNEY IN DIABETIC KIDNEY DISEASE

DKD was first described in 1936 by the presence of mesangial expansion and nodular glomerulosclerosis in a classic paper by Kimmelstiel and Wilson.³⁴⁷ Structural

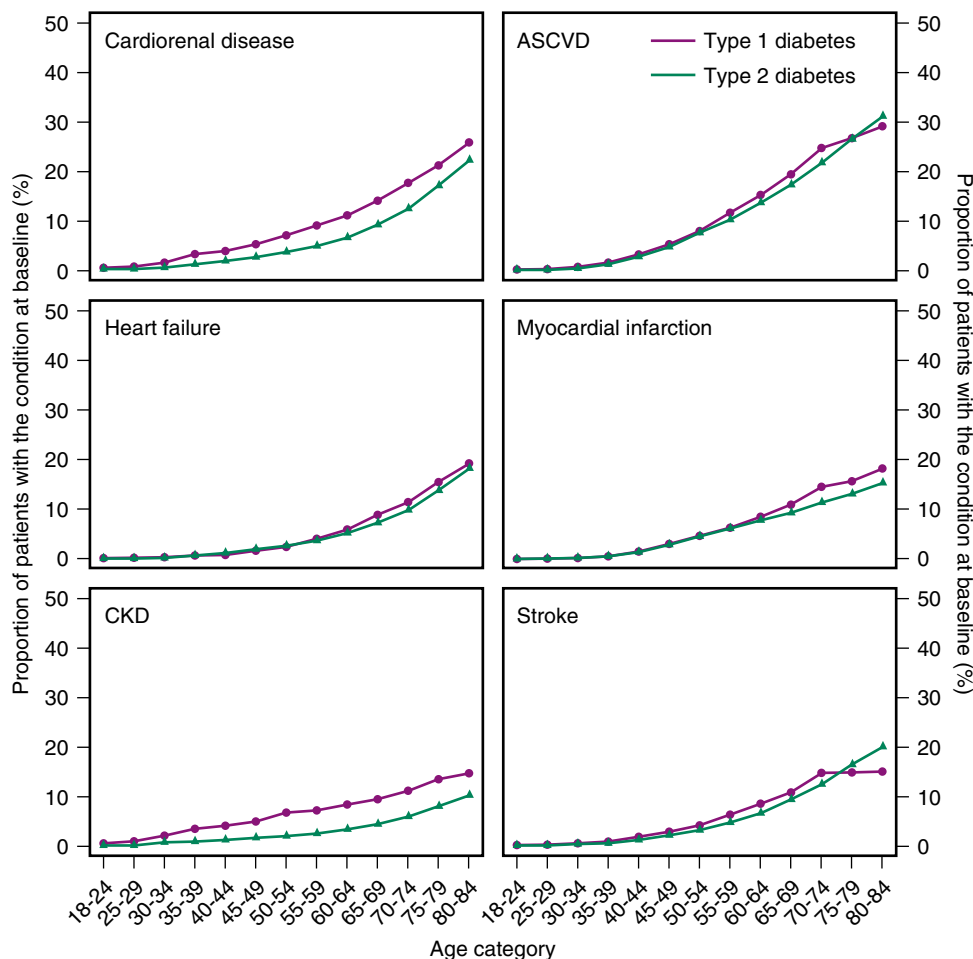


Fig. 41.5 Rates of cardiovascular and kidney disease in type 1 and type 2 diabetes.³³³ Age-stratified baseline prevalence of chronic kidney disease (CKD), heart failure (HF or CKD), stroke, myocardial infarction (MI), and atherosclerotic cardiovascular disease (ASCVD) (MI or stroke).

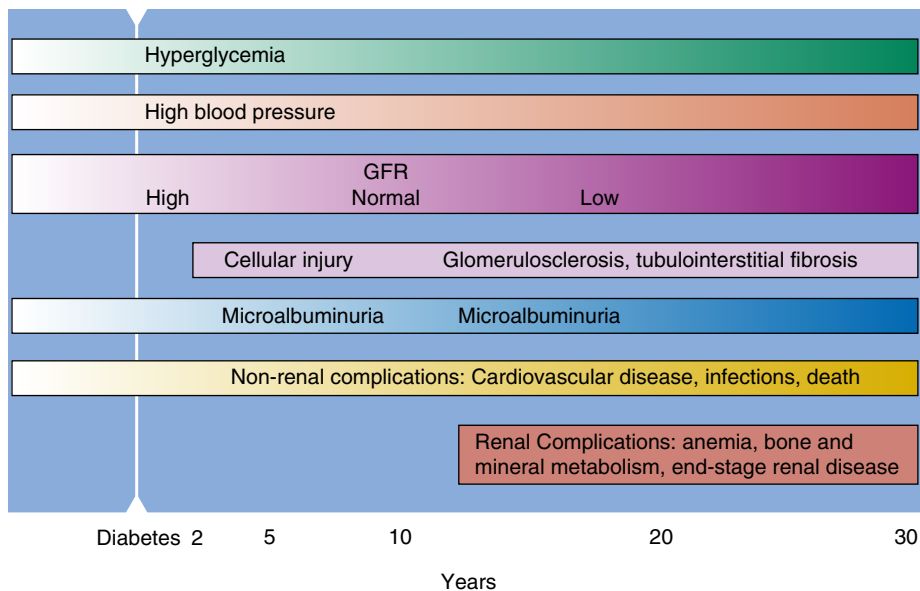


Fig. 41.6 Conceptual model of the natural history of diabetic kidney disease.³⁴⁵ Duration of diabetes, in years, is presented on the horizontal axis. Timeline is well characterized for type 1 diabetes mellitus; for type 2 diabetes mellitus, timeline may depart from the illustration due to the variable timing of the onset of hyperglycemia. Kidney complications: anemia, bone and mineral metabolism, retinopathy, and neuropathy.

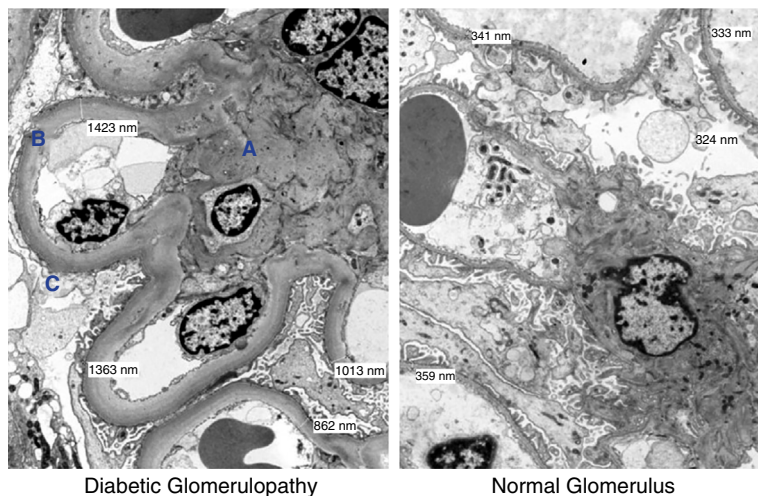


Fig. 41.7 Electron microscope images of structural changes in diabetic kidney disease.³⁴⁵ Structural changes in diabetic glomerulopathy found with electron microscopy. *A* indicates marked expansion of the mesangium. *B* indicates marked diffuse thickening of capillary basement membranes (to three times the normal thickness in this case). *C* indicates segmental effacement of the visceral epithelial foot processes. Original magnification, $\times 3500$.

changes of DKD are similar in both types of diabetes and occur within multiple kidney compartments.³⁴⁸ The earliest changes are enlargement of the kidney due to glomerular and tubular hypertrophy that develops in response to a number of stimuli including oxidative stress, enhanced filtration, tubular expression of several growth factors (e.g., insulin-like growth factor I [IGF1], platelet-derived growth factor [PDGF], VEGF and EGF) and activation of several inflammatory pathways (e.g., JAK-STAT pathway and mTORC1).³⁴⁹ Thickening of the glomerular basement membrane (GBM) becomes evident within 1.5 to 2 years after diagnosis of type 1 diabetes and is associated with increased densities of $\alpha 3$ and $\alpha 4$ chains of type IV collagen.^{60,350-356} Another early finding is a thickening of tubular basement membrane (TBM) that takes place in parallel with thickening of GBM.^{63,354,357} Additional features of early DKD include expansion of glomerular mesangium that can be detected by light microscopy as early as 4 to 5 years after the onset of type 1 diabetes (Fig. 41.7).^{345,354,358} Mesangial expansion is due to increased mesangial matrix and, to a lesser extent, increases in mesangial cell

cellularity.³⁵⁹ It is associated with protrusion of mesangium into peripheral capillary walls, leading to reduced filtration surface area and an inverse relationship between $V_v(\text{Mes/glom})$ and peripheral GBM filtration surface density, $S_v(\text{PGBM/glom})$.^{122,360} In part due to an adaptive response to the thickening GBM, podocytes hypertrophy, flatten, and detach. Eventually, severe loss of podocytes ensues, leaving large subepithelial surfaces of the GBM denuded (Fig. 41.8).^{353,354,361-364} Progression of the structural glomerular changes includes thickening of the Bowman capsule and mesangial matrix expansion that in later stages develops into a form of nodular glomerulosclerosis known as Kimmelstiel-Wilson nodules.^{359,365,366} Accumulation of plasma proteins, or hyaline material, may form between the glomerular endothelium and the GBM and between the glomerular tuft and Bowman capsule.^{345,365,367,368} Dissection of the glomerulotubular junction can lead to stricture and occlusion of the glomerular tubular outlet and eventually create atubular glomeruli (Fig. 41.9).^{369,370} The tubulointerstitium progressively develops fibrosis and inflammation, following early findings of thickening of

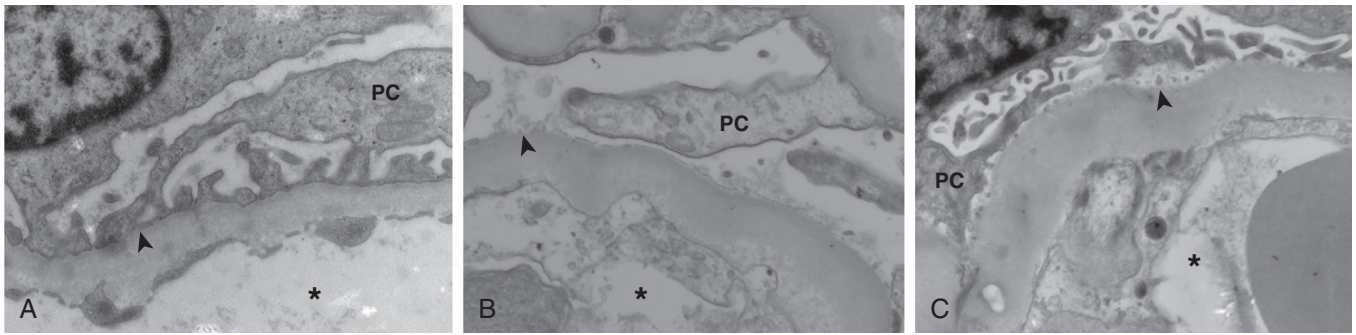


Fig. 41.8 Detachment of podocytes (PC) from the glomerular basement membrane in diabetic kidney disease (DKD).³⁶⁴ (A) Glomerular basement membrane covered by intact foot processes in mild DKD. With advancement of DKD, some areas of glomerular basement membranes with complete (B), or partial (C) detachment of podocytes are observable. *Capillary lumen. Arrowhead shows the podocyte aspect of the glomerular basement membrane.

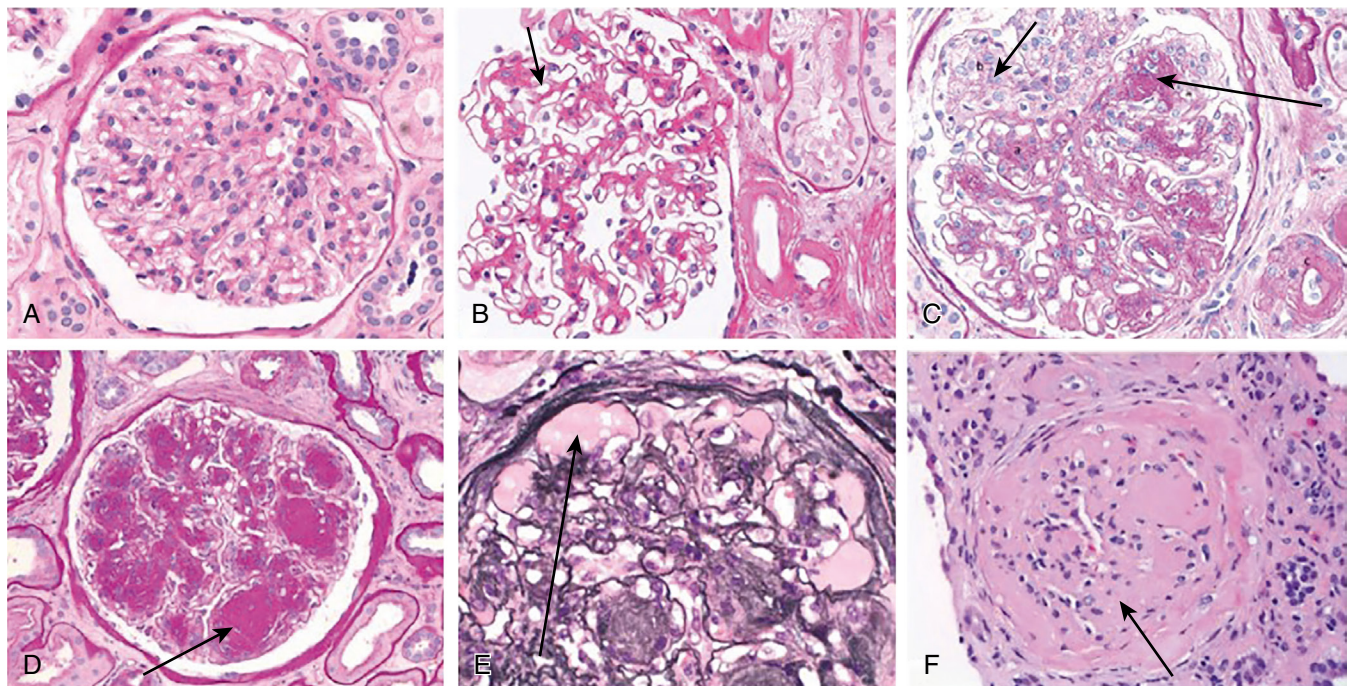


Fig. 41.9 Glomerular changes in diabetic kidney disease.³⁴⁵ Changes in glomerular histology in diabetic glomerulopathy. (A) Normal glomerulus. (B) Diffuse mesangial expansion with mesangial cell proliferation. (C) Prominent mesangial expansion with early nodularity and mesangiolysis. (D) Accumulation of mesangial matrix forming Kimmelstiel-Wilson nodules. (E) Dilation of capillaries forming microaneurysms, with subintimal hyaline (plasmatic insudation). (F) Obsolescent glomerulus. A–D and F were stained with period acid–Schiff stain, and E was stained with Jones stain. Original magnification, $\times 400$.

the tubular basement membranes, culminating in tubular atrophy.³⁶³ With disease progression, efferent arterioles and capillary loops become hyalinized. In small arterial branches, arterioles, and glomerular capillaries there is evidence of exudative lesions formed by accumulation plasma proteins in the subendothelium. Exudative lesions present as periodic acid–Schiff–positive and electron-dense deposits on microcopy and electron microscopy, respectively, and may result in luminal compromise (e.g., hyaline arteriosclerosis). Changes in arteriosclerosis and hyalinosis of the blood vessels are particularly present at the vascular pole. Similar subepithelial deposits are seen in the Bowman capsule (“capsular drop lesion”) and proximal

renal tubules. The tubulointerstitial damage progresses in concert with glomerulosclerosis. In advanced stages, coalescence of segmental and global glomerular sclerosis with severe interstitial fibrosis and tubular atrophy is evident (Fig. 41.10).^{345,368} The international Renal Pathology Society work group developed a histopathologic classification system to describe and grade the severity of DKD (Tables 41.3 and 41.4).³⁷¹

Structural changes in DKD are similar in type 1 and type 2 diabetes, but the pathologic findings and correlations with the clinical presentation are less predictable in type 2 diabetes.³⁷² For instance, an autopsy study of predominantly type 2 diabetes (90%) patients found that $>20\%$ (20/106)

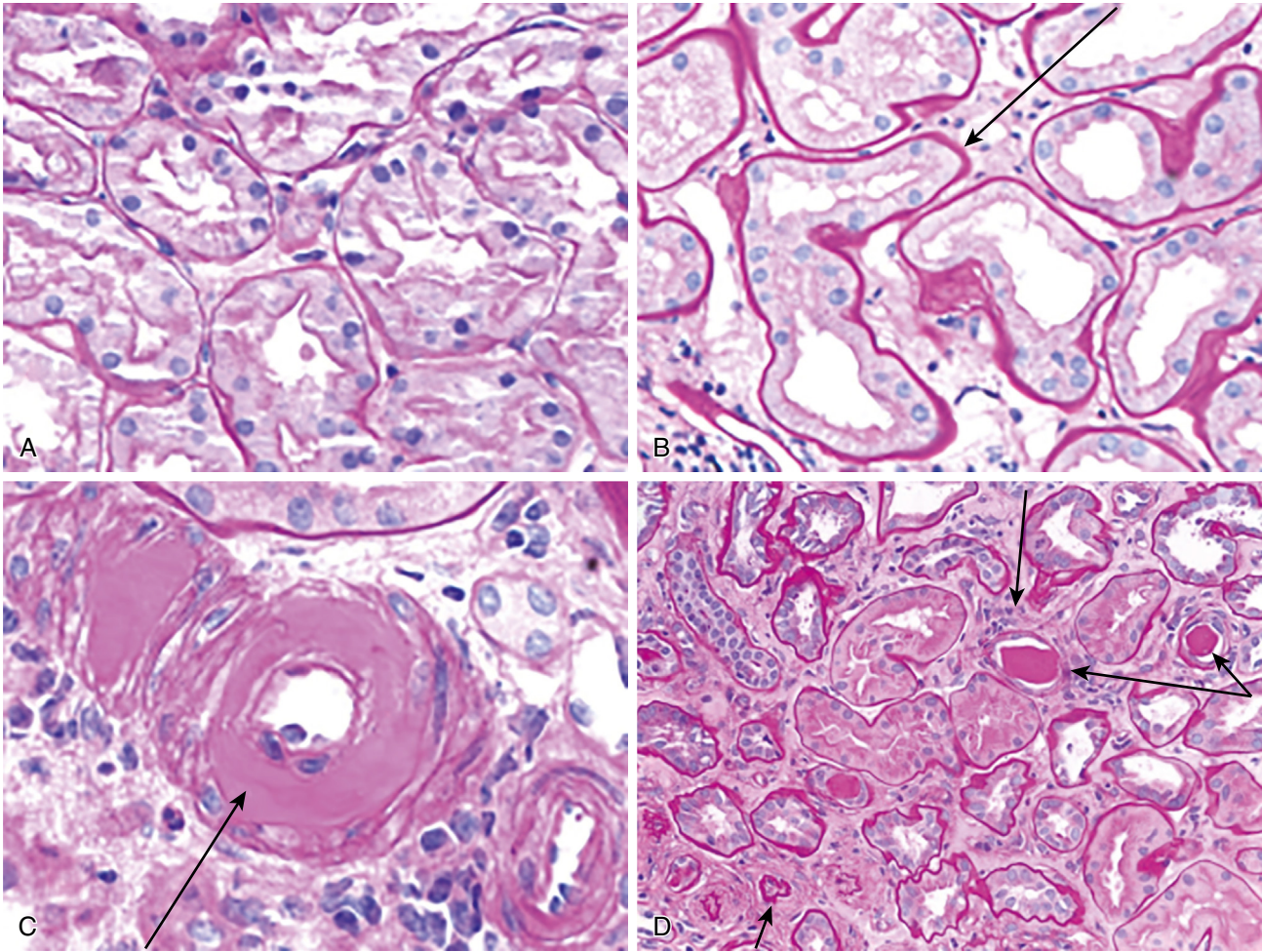


Fig. 41.10 Tubulointerstitial changes and arterial hyalinosis in diabetic kidney disease.³⁴⁵ Tubulointerstitial changes in diabetic kidney disease. (A) Normal renal cortex. (B) Thickened tubular basement membranes and interstitial widening. (C) Arteriole with an intimal accumulation of hyaline material with significant luminal compromise. (D) Renal tubules and interstitium in advancing diabetic kidney disease, with thickening and wrinkled tubular basement membranes (*solid arrows*), atrophic tubules (*dashed arrows*), some containing casts, and interstitial widening with fibrosis and inflammatory cells (*dotted arrow*). All sections were stained with period acid–Schiff stain. Original magnification, $\times 200$.

Table 41.3 Classification of Diabetic Kidney Disease Glomerular Lesions According to Tervaert et al³⁷¹

Class	Description	Inclusion criteria
I	Mild or nonspecific LM changes and EM-proven GBM thickening	Biopsy does not meet any of the criteria mentioned below for class II, III, or IV GBM >395 nm in female and >430 nm in male individuals 9 years of age and older
IIa	Mild mesangial expansion	Biopsy does not meet criteria for class III or IV Mild mesangial expansion in $>25\%$ of the observed mesangium
IIb	Severe mesangial expansion	Biopsy does not meet criteria for class III or IV Severe mesangial expansion in $>25\%$ of the observed mesangium
III	Nodular sclerosis (Kimmelstiel–Wilson lesion)	Biopsy does not meet criteria for class IV At least one convincing Kimmelstiel–Wilson lesion
IV	Advanced diabetic glomerulosclerosis	Global glomerular sclerosis in $>50\%$ of glomeruli Lesions from classes I through III

^aOn the basis of direct measurement of GBM width by EM, these individual cutoff levels may be considered indicative when other GBM measurements are used.

EM, Electron microscopy; GBM, glioblastoma multiforme; LM, light microscopy.

Table 41.4 Classification of Diabetic Kidney Disease Interstitial and Vascular Lesions According to Tervaert et al³⁷¹

Lesion	Criteria	Score
Interstitial lesions		
IFTA	No IFTA	0
	<25%	1
	25%-50%	2
	>50%	3
Interstitial inflammation	Absent	0
	Infiltration only in relation to IFTA	1
	Infiltration in areas without IFTA	2
Vascular lesions		
Arteriolar hyalinosis	Absent	0
	At least 1 area of arteriolar hyalinosis	1
	More than 1 area of arteriolar hyalinosis	2
Presence of large vessels	—	Yes/no
Arteriosclerosis (score worst artery)	No intimal thickening	0
	Intimal thickening less than thickness of media	1
	Intimal thickening greater than thickness of media	2

IFTA, Interstitial fibrosis and tubular atrophy.

did not have a clinical diagnosis of DKD while alive and, specifically, did not have albuminuria.³⁷³ Structural changes in this cohort encompassed the spectrum of histopathologic classes of DKD.³⁷³ These findings were confirmed in diabetic kidney transplant donors (98% type 2 diabetes) in whom structural changes preceded clinical evidence of DKD.³⁷⁴

Clinical Relevance: Pathology

The natural history of DKD starts with an initial long clinically silent period during which albumin excretion rate remains normal and GFR is normal or high, while DKD lesions slowly progress. Therefore albuminuria is not a sensitive marker to detect early DKD. On the other hand, when patients develop microalbuminuria and, especially, macroalbuminuria, DKD lesions are typically far advanced and GFR loss becomes accelerated. Kidney biopsy provides important insight about severity and prognosis of DKD. However, predictive value of biopsy findings in early DKD, when there is more opportunity to rescue the kidney, for later progression to kidney failure remains limited. This calls for better methods to identify patients at high risk. Studies combining clinical information with structural data and biomarker discovery are needed to achieve this goal.

PATHOPHYSIOLOGY

The pathogenesis of DKD is multifactorial in origin, involving a variety of metabolic and hemodynamic factors that, combined with certain susceptibility genes, can affect different target cells in the kidney.³⁷⁵ Metabolic changes that occur with the onset of diabetes substantially alter kidney physiology in ways that promote inflammation and fibrosis and ultimately lead to kidney failure. Early perturbations in response to hyperglycemia include changes in hemodynamic function such as glomerular hyperfiltration and the selective behavior of the glomerular capillary wall to macromolecules. Glomerular endothelial cells, epithelial, mesangial, and tubular cells can be differently affected by key elements of the diabetic milieu, and crosstalk among different cell types and different organs affected by diabetes is suggested by a large variety of experimental and clinical data. Irrespective of the initiating factor and of the target cells, several key pathways modulated in DKD have been identified (Fig. 41.11).

SELECTIVE GLOMERULAR PERMEABILITY

The excretion of albumin in the urine is determined by the amount of albumin filtered across the glomerular capillary barrier and the amount reabsorbed by the tubular cells. The glomerular capillary wall serves as a filter that discriminates among molecules on the basis of size, electrical charge, and configuration. Studies of glomerular filtrate collected by micropuncture or narrow size fractioning of exogenous polymers, such as dextran, indicate that albuminuria is primarily the result of impairment of the electrostatic barrier within the glomerulus, consequent to a decrease in endothelial cell glycocalyx³⁷⁶⁻³⁷⁸ and heparan sulfate content of the glomerular basement membrane,³⁷⁹ as well as by changes in the size-selective properties of the glomerular capillary barrier.³⁸⁰⁻³⁸⁶ The size-selective defect in the glomerular capillary appears to become more prominent with higher levels of albumin excretion among patients with either type 1 or type 2 diabetes.^{385,387,388} Morphometric data in kidney tissue from Pima Indians with macroalbuminuria demonstrate a significant correlation between the magnitude of this permselective defect and podocyte foot-process width, which is not observed in those with microalbuminuria.³⁸⁸ These findings are consistent with the view that permselective defects responsible for increased albumin excretion may be focal and are likely due to podocyte foot-process effacement and simplification, as well as to defective intercellular junctions.³⁸⁹

Multiphoton fluorescence techniques permit direct imaging of the structure and function of living kidney tissue.³⁹⁰ In some studies, this technique revealed what appeared to be a much higher albumin glomerular sieving coefficient than was calculated or measured by micropuncture, prompting some investigators to propose alternative explanations for the facilitated urine clearance of albumin in DKD,^{390,391} including the idea that albuminuria is the result of proximal tubular cell dysfunction in retrieving and degrading albumin.^{392,393} Other studies using improved imaging techniques do not support this concept and confirm that glomerular filtration barrier permeability to macromolecules is largely restricted to areas of podocyte damage.³⁹⁴⁻³⁹⁶

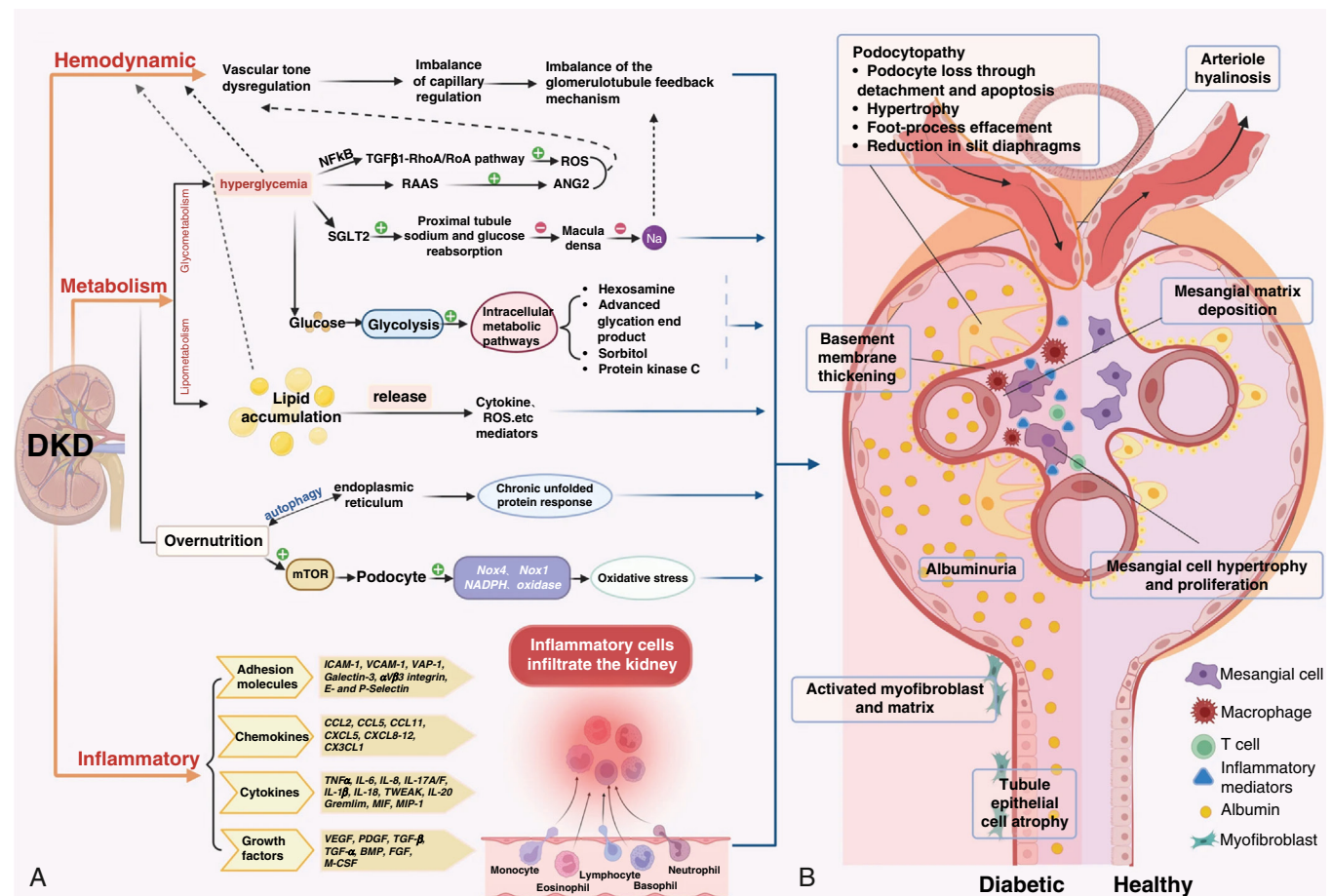


Fig. 41.11 Pathology of the glomerulus and tubules in diabetic kidney disease (DKD).³⁷⁵ (A) Classical pathologic mechanisms of DKD. It mainly includes hemodynamic, metabolic disturbances, and inflammation, which often interact with each other. 1. Hemodynamic disturbances lead to dysregulation of tubulobulbar feedback balance. 2. Metabolic disorders are crucial to the pathogenesis of DKD. Hyperglycemia affects pathways such as TGFβ1-RhoA/RoA pathway, RAAS, proximal tubular sodium and glucose reabsorption, and intracellular metabolism; abnormal lipid metabolism can affect the release of mediators such as cytokines and ROS; in the presence of nutrient overload in the organism, endoplasmic reticulum autophagy leads to a chronic unfolded protein response, and mTOR also disturbs the podocytes leading to oxidative stress. 3. Inflammation promotes the release of inflammatory mediators such as adhesion molecules, chemokines, cytokines, and growth factors, causing renal infiltration of inflammatory cells. (B) Schematic representation of the pathologic damage of DKD. Differences in structural changes of glomeruli and tubules in the diabetic setting and in the healthy state. Diabetic glomerulopathy is characterized by arterial hyalinization, thylakoid stromal deposition, basement membrane thickening, glomerular thylakoid cell hypertrophy and proliferation, podocytopeny, proteinuria, tubular epithelial atrophy, activated myofibroblasts, and stromal accumulation. *ANG2*, Angiotensin II; *BMP*, bone morphogenetic protein; *CCL*, CC chemokine ligand; *CXCL*, C-X-C motif chemokine ligand; *FGF*, fibroblast growth factor; *ICAM-1*, intercellular cell adhesion molecule-1; *IL*, interleukin; *M-CSF*, macrophage colony-stimulating factor; *MIF*, macrophage migration inhibitory factor; *MIP-1*, macrophage inflammatory protein-1; *mTOR*, mammalian target of rapamycin; *NADPH*, nicotinamide adenine dinucleotide phosphate; *NFκB*, nuclear factor κ-B; *NOX*, NADPH oxidase; *PDGF*, platelet-derived growth factor; *RAAS*, renin-angiotensin-aldosterone system; *ROS*, reactive oxygen species; *SGLT2*, sodium-dependent glucose transporters 2; *TGF-β*, transforming growth factor-β; *TNF*, tumor necrosis factor; *TWEAK*, tumor necrosis factor-like weak inducer of apoptosis; *VAP-1*, vascular adhesion protein-1; *VCAM-1*, vascular cell adhesion molecule-1; *VEGF*, vascular endothelial growth factor.

GLOMERULAR HEMODYNAMIC FUNCTION

In healthy adults, the GFR ranges from 90 to 120 mL/min/1.73 m², is stable through midadult life, and declines by approximately 1 mL/min per year after 50 years of age.³⁹⁷⁻³⁹⁹ The onset of diabetes is associated with hemodynamic changes in the glomerular circulation that lead to increased renal plasma flow, glomerular capillary hyperperfusion, and an increased glomerular transcapillary hydraulic pressure gradient, which together contribute to an increase in GFR.⁴⁰⁰⁻⁴⁰⁵ The prevalence of glomerular hyperfiltration, frequently defined as a GFR of at least two standard deviations above the mean GFR in

persons with normal glucose tolerance, varies from 10% to 67% in persons with type 1 diabetes and from 6% to 73% in those with type 2 diabetes (Fig. 41.12).³²⁴ Large variations in these estimates are attributed to differences in age, race and ethnicity, glycemic control, duration of diabetes, absence of diet standardization, the definition of hyperfiltration, and methodologies used to measure and report GFR among different populations. Glomerular capillary hypertension and the ensuing increase in filtration pressure are partly responsible for the elevation of GFR, but various other glomerular and tubular factors also influence the magnitude of the hyperfiltration.⁴⁰⁶⁻⁴⁰⁸

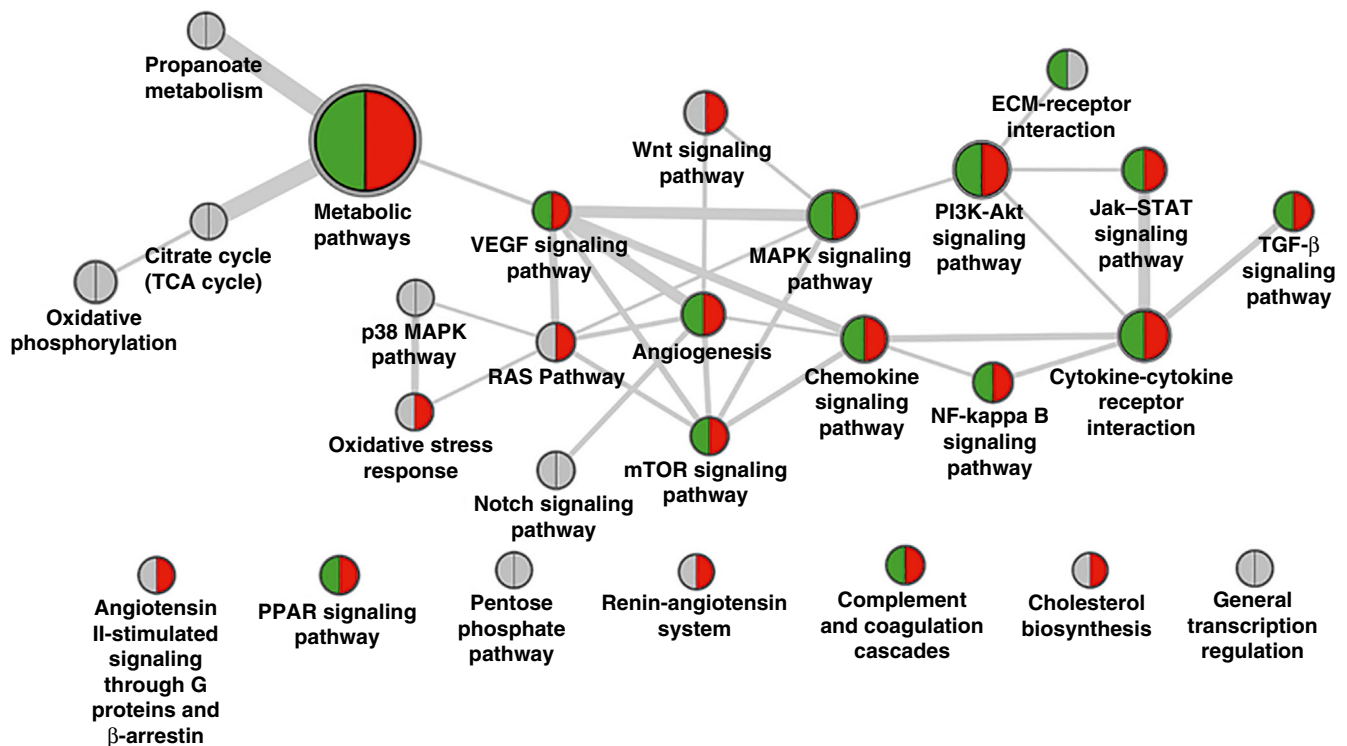


Fig. 41.12 Schematic (net) effect of factors implicated in the pathogenesis of glomerular hyperfiltration in diabetes. Several vascular and tubular factors^{32,48,123–126} are suggested to result in a net reduction in afferent arteriolar resistance, thereby increasing (single-nephron) glomerular filtration rate (GFR). Effects of insulin per se seem to depend on insulin sensitivity.^{96,97} A net increase in efferent arteriolar resistance—leading to increased GFR—is proposed for other vascular factors.^{32,42,71,124,127} Growth hormone¹²⁸ and insulin-like growth factor¹²⁹ likely increase filtration by augmenting total renal blood flow, without specific arteriolar preference. Glucagon and vasopressin seem to (principally) act through transforming growth factor (TGF).⁴⁸ Intrinsic defects of electromechanical coupling or alterations in signal transduction in afferent arterioles may impair vasoactive responses to renal hemodynamic (auto) regulation.³² Augmented filtration by increases in the ultrafiltration coefficient, and net filtration pressure via reduction in intratubular volume and subsequent hydraulic pressure in the Bowman space are not depicted. Several vascular factors may be released or activated after a (high-protein) meal (e.g., nitric oxide, cyclooxygenase-2 prostanoids, and angiotensin II),^{48,50,130} whereas TGF becomes (further) inhibited, through increased amino acid- (and glucose) coupled sodium reabsorption in the proximal tubule^{49,50} and/or increased glucagon/vasopressin-dependent sodium reabsorption in the thick ascending limb.⁴⁸ These changes may collectively play a part in postprandial hyperfiltration. COX-2, Cyclooxygenase-2; ETA, endothelin A receptor.

After the initial elevation at the onset of diabetes, GFR decreases in response to glycemic control in both type 1 and type 2 diabetes^{409–411} but usually not to levels found in nondiabetic persons.^{403,412} In some patients, this reversal could reflect the initiation of progressive kidney disease, as suggested by the appearance of global glomerular sclerosis and the fall in single-nephron filtration coefficient.⁴¹³ In others, it could represent a purely functional change in kidney vasomotion associated with improvement in glycemic control or simply the intrinsic variability of GFR in the absence of significant histopathologic changes.⁴¹⁴ Distinguishing between these two potential causes of GFR decline requires either observation of GFR over an extended period to determine whether it plateaus, reverses direction, or continues to decline to pathologic levels or attention to other biomarkers including albuminuria.⁴¹⁵ Nevertheless, differentiating between these possibilities is crucial because progressive kidney function decline indicative of underlying pathology may precede the onset of albuminuria in both type 1 and type 2 diabetes.^{416–420} This progressive decline is predominantly linear, but the rate of decline varies widely among individuals. Importantly, half of those in a type 1

diabetes cohort who developed kidney functional decline progressed from a normal eGFR to kidney failure within 2 to 10 years.

PRIMARY CELL TARGETS

GLOMERULAR ENDOTHELIAL CELLS

The role of the glomerular endothelial cells (GECs) and the thickness of the glycocalyx as one of the early targets in DKD has generated much attention. Both a reduction in the thickness of the glycocalyx and a reduction in the fenestration of the endothelium have been described as early features of clinical DKD. GEC injury can occur via hemodynamic stimuli that cause reduced nitric oxide (NO) bioavailability via suppression of endothelial nitric oxide synthase (eNOS), or it can result from growth factor–driven altered metabolism. Triglyceride-rich lipoproteins, advanced glycation end products, and inflammatory mediators can all negatively influence the thickness of the glycocalyx.⁴²¹ Several interventions are effective in restoring the glycocalyx barrier and protecting GECs including atrasentan,⁴²² C3a and C5a receptor antagonism,⁴²³ and VEGF-A165b

administration via VEGFR2 phosphorylation.⁴²⁴ Among drug targets, glomerular endothelin-1 receptor type A (EdnRA) is activated in DKD and linked to mitochondrial dysfunction.⁴²⁵ The glomerular endothelium can also be the target of a “gut to kidney” connection, as the GLP1 receptor agonist exendin-4 ameliorates lipotoxicity-induced GECs injury by improving ATP-binding cassette transporter A1 (ABCA1)-dependent cholesterol efflux.⁴²⁶ High glucose is sufficient to cause local upregulation of the renin-angiotensin system in GECs in association with increased fenestration and decreased glycosaminoglycans content, a phenotype that can be prevented by ARB treatment.⁴²⁷

The role of angiotensins is potentially important. Under physiological conditions, they regulate the integrity of the glomerular filtration barrier, but when dysregulated in diabetes they may be sufficient to drive the pathobiology of DKD.⁴²⁸ These findings may lead to the development of targeted glomerular angiotensin-1 therapies.^{428,429}

As GECs are also the first cell being encountered by any circulating stimulus relevant to diabetes, they can be a direct target of the diabetes milieu, but they can also serve as a cell sending paracrine signals to adjacent mesangial cells and podocytes. Many studies suggest the presence of crosstalk between GEC and mesangial cells (primarily platelet-derived growth factor [PDGF]-B/PDGF β mediated) or GEC and podocytes (primarily mediated by KLF2, angiotensins, VEGFA, FLK1, FLT1, TIE2, and endothelin-1).^{430,431} The possibility of an endothelial-to-mesenchymal transformation has also been suggested.⁴³²

Increasing attention has been paid to bioactive sphingolipids including sphingosine-1-phosphate (S1P) and the role of S1P receptor (S1PR) pathway activation in DKD. S1P is a bioactive sphingolipid generated from sphingosine through phosphorylation by sphingosine kinase (SPHK). It regulates distinct biological effects via activation of different S1PRs, S1PR1-5, on target cells including glomerular cells. This pathway was found to be upregulated in podocytes and endothelial and mesangial cells. Increased S1P receptor 2 (S1PR2) expression was found in kidneys of 28 patients with biopsy-confirmed DKD when compared with controls. Thereby, endothelial S1PR2 activation is thought to induce endothelial-to-mesenchymal transformation via the Wnt3a/RhoA/ROCK1/ β -catenin pathway, leading to impairment of the endothelial barrier function.⁴³³ In further support of a link between the activation of S1P/S1PR2 signaling and endothelial barrier dysfunction in DKD, SPHK1 expression is increased in kidney sections of patients with DKD when compared with kidney sections of healthy subjects and patients with kidney cancer,⁴³⁴ and S1P-mediated S1PR2 activation in HUVECs was found to cause endothelial cell barrier dysfunction.⁴³⁵

PODOCYTES

An early and central event in the development of DKD is the loss of podocytes from the glomerulus.⁴³⁵⁻⁴³⁹ Podocyteopenia correlates with DKD progression^{438,440} and is inversely related to the control of hypertension and diabetes,⁴³⁹ but it can also occur independently of blood pressure.⁴⁴¹ Several signaling pathways are implicated in podocyte injury in diabetes. These include loss or redistribution of slit diaphragm proteins such as nephrin, altered insulin receptor signaling, altered nutrient sensing via the mechanistic target

of rapamycin (mTOR) pathway, and reactivation of key developmental pathways such as the Notch pathway and the Wnt/ β -Catenin pathway.

Hyperglycemia is the foundation for direct podocyte injury in DKD through multiple pathways including the polyol pathway, formation of advanced glycation end products, modulation of protein kinase C (PKC), and hexosamine pathway.⁴⁴² However, podocyte function is also modulated by a variety of circulating factors produced by other organs, suggesting a complex crosstalk between the kidney and other organs in diabetes. Podocytes can be the direct target of traditional components of the renin-angiotensin system, including angiotensin II,⁴⁴³⁻⁴⁴⁵ aldosterone,^{446,447} and prorenin.^{448,449} The importance of the local renin-angiotensin system in the development of proteinuria comes from a mouse model overexpressing angiotensin II type 1 receptor (AT1R) in podocytes.⁴⁴⁴ Renin-angiotensin system blockade may also affect podocyte function through the systemic modulation of adipokines and insulin sensitivity.⁴⁵⁰ While the role of locally produced ACE requires further investigation, clinical and experimental evidence suggests kidney protection by aldosterone blockade.⁴⁵¹⁻⁴⁵⁴ Although both systemic and local effects may be involved, mineralocorticoid receptor (MR) activation in podocytes alters the filtration barrier and results in proteinuria.⁴⁴⁶ Aldosterone-independent MR ligands that are relevant to podocyte function, such as the small GTPase Rac1, have been identified, underlying the complexity of podocyte signaling in DKD.⁴⁴⁷ Furthermore, insulin,^{455,456} insulin-like growth factors (IGFs),⁴⁵⁷ adiponectin,⁴⁵⁸ sex hormones,⁴⁵⁹ growth hormone,⁴⁶⁰⁻⁴⁶² adrenocorticotrophic hormone (ACTH),⁴⁶³ growth hormone-releasing hormone (GHRH),⁴⁶⁴ thyroid hormone,⁴⁶⁵ and vitamin D⁴⁶⁶ are involved in direct modulation of podocyte function.

Podocyte-specific deletion of the insulin receptor in mice is sufficient to cause a phenotype similar to DKD in the absence of hyperglycemia, strongly supporting the importance of insulin signaling in the preservation of podocyte function.^{467,468} This is supported by clinical studies, as IR correlates with the development of albuminuria in patients with either type 1 or type 2 diabetes,^{74,78,79,469} their siblings,^{76,78} and nondiabetic individuals.⁴⁷⁰ Furthermore, insulin-sensitizing agents of the class of thiazolidinediones offer a degree of kidney protection in patients with DKD,⁴⁷¹ as well as in insulinopenic experimental animal models of DKD.⁴⁷² Experimental data support this clinical observation, as IR occurs *in vitro*⁴⁵⁶ and *in vivo*⁴⁷³ at early stages of DKD and may occur through PKC- β -induced insulin receptor substrate-1 dysfunction. Genetic manipulation of several elements of the insulin signaling cascade other than the insulin receptor alters the course of DKD in mice and affects relevant nutrient sensing and prosurvival pathways.⁴⁷⁴ Both glucose transporter 1 and glucose transporter 4 expression can contribute to the pathogenesis of DKD.⁴⁷⁵⁻⁴⁷⁷ In addition, several mediators of glucose transporter 4 trafficking in podocytes are linked to podocyte function in DKD. These include septin 7,⁴⁷⁸ CD2-associated protein,^{479,480} nucleobindin-2,⁴⁸¹ Synip,⁴⁸² insulin receptor substrate 2, and phosphatase and tensin homolog.⁴⁸³ Epigenetic changes in the promoter of Src homology region 2 domain-containing

phosphatase-1 may also have a role in persistence of IR in podocytes in the diabetes state.⁴⁸⁴

Podocyte IR has also been linked to mitochondrial function, endoplasmic reticulum stress, and activation of the unfolded protein response.⁴⁸⁵ Autophagy is essential to maintain podocyte homeostasis, as haploinsufficiency of mTORC1 activates autophagy and protects from DKD while mTORC1 activation leads to acceleration of DKD.^{486,487} While an increase in reactive oxygen species drives the vascular complications of diabetes, some data suggest that increased mitochondrial superoxide production may indicate healthier mitochondria, thus opening a theory of mitochondrial hormesis.⁴⁸⁸ In response to excess glucose exposure or nutrient stress, the consequent reduction in mitochondrial oxidative phosphorylation may lead to the release of nonmitochondrial oxidants and to a proinflammatory and profibrotic response. Restoration of superoxide production with activation of AMP-activated protein kinase (AMPK) may promote organ healing.⁴⁸⁹

Besides traditional hormones, additional important modulators of podocytes function include free fatty acids (FFAs),⁴⁹⁰ lipoproteins,^{491,492} and angiopoietins, and more studies are needed to support their specific function in DKD before translation into new therapeutic strategies. Podocytes are also the main source of VEGFA production, with both excess and defective production of VEGFA contributing to worsening DKD⁴⁹³ through its role as a key element in the paracrine crosstalk between podocytes and endothelial cells,⁴⁹⁴⁻⁴⁹⁶ and more recently through tubulovascular crosstalk in the peritubular microvasculature of the kidney.⁴⁹⁶ Besides the paracrine functions of VEGF-A, an autocrine function in podocytes has also been described.⁴⁹⁷ VEGF-A is a key modulator of slit diaphragm protein interactions⁴⁹⁸ and the actin cytoskeleton. REF nodular glomerulosclerosis resembling DKD has been described when either VEGF-A⁴⁹⁹⁻⁵⁰¹ or Vegf164⁵⁰² are overexpressed in podocytes in eNOS null mice or in models of type 1 diabetes. Therapeutic strategies targeting this system have not been successful, although development of soluble fms-related tyrosine kinase 1 (sFLT1) decoy peptides may stabilize podocytes through binding to a podocyte-specific GM3 ganglioside, thus increasing cell adhesion and actin remodeling.⁵⁰³

Crosstalk between the adipose tissue and kidney has also been invoked. Podocytes express both adiponectin receptors 1 and 2 (AR1 and AR2)⁴⁵⁸ and adiponectin null mice develop proteinuria and podocyte injury that can be reversed by the administration of recombinant adiponectin, which modulates podocyte oxidative stress through an AMPK-dependent modulation of NOx 4 in podocytes. A cause-effect relationship between adiponectin and albuminuria has been reported,⁴⁵⁸ suggesting a protective effect of adiponectin on podocyte function and albuminuria. Serum adiponectin levels are elevated in patients with type 1 diabetes and type 2 diabetes who have DKD.⁵⁰⁴⁻⁵⁰⁸ Conversely, low plasma adiponectin was found to predict the progression of kidney disease in another cohort of patients with type 2 diabetes.⁵⁰⁹ While adiponectin regulates lipid metabolism via AR1-mediated induction of AMPK phosphorylation and AR2-mediated increased peroxisome proliferative-activated receptor (PPAR)- α expression, leading to increased fatty acid

oxidation, it also has functions that do not involve fatty acid metabolism. For example, adiponectin binding to ARs can modulate sphingolipid metabolism by activating ceramidase, the enzyme that converts cell toxic ceramide to S1P.^{510,511} A role for low adiponectin levels in DKD and in the conversion of ceramide to S1P is further supported by a dominant negative protein truncating ADIPOQ mutation in a multigenerational family. Carriers of the mutation present with diabetes, reduced circulating adiponectin, increased long-chain ceramide levels, and DKD. The identification of this novel mutation in the *ADIPOQ* gene and the fact that several single nucleotide polymorphisms (SNPs) in the *ADIPOQ* gene, including the promoter, are associated with increased risk of the development of type 2 diabetes and DKD suggests it may be a clinically relevant pathway.⁵¹²

Epidemiologic data suggest that estrogens may provide a protective effect against the development and progression of DKD that is lost after menopause,⁵¹³⁻⁵¹⁵ and treatment with 17 β -estradiol protects from experimental DKD.^{516,517} Circulating 17 β estradiol affects the podocytes expression of estrogen receptor beta, which is involved in the modulation of extracellular matrix production via matrix metalloproteinases.⁴⁵⁹

Kidney expression and circulating concentrations of an antiaging hormone, Klotho, are reduced in DKD,⁵¹⁸⁻⁵²⁰ and Klotho replacement may ameliorate proteinuria by targeting transient receptor potential cation channel, subfamily C, member 6 channels in podocytes,⁵²¹ and rescue diabetic mice from glomerular injury.⁵²² Altered growth hormone (GH)/insulin-like growth factor (IGF)-1 axis has been described in diabetes and DKD.⁵²³ Several studies suggest a direct relationship between the activity of the GH/IGF-1 axis and certain features of DKD, such as hyperfiltration and microalbuminuria,⁵²⁴ and somatostatin analogs may be kidney protective in diabetes.⁵²⁵ In experimental models, overexpression of GH results in severe glomerulosclerosis,⁵²⁶ and inhibition of GH action through different mechanisms may improve DKD.^{461,462,527} Interestingly, podocytes express GH receptor, and signaling through the GH receptor in podocytes affects oxidative stress and actin remodeling, two important features of podocyte function.⁴⁶⁰ Taken together, these experimental data suggest that medicines targeting the GH receptor or GH signaling cascade may represent an approach to prevent or treat DKD. High expression of GH-releasing hormone has been detected in the kidney,⁴⁶⁴ but its specific function in the kidney remains to be established.

Podocytes are also the direct target of profibrotic molecules such as TGF- β , which binds to receptors on podocytes and activates a SMAD7-mediated signaling pathway leading to aberrant extracellular matrix production.⁵²⁸ However, antifibrotic agents have consistently failed to protect from DKD, strongly suggesting that fibrosis is a downstream outcome.⁵²⁹ As TGF- β may also trigger many epigenetic changes leading to fibrosis, additional studies are required to investigate targeting the fibrotic pathways via this mechanism.

TNF and nuclear factor of activated T cell (NFAT)-mediated impairment of ABCA1-dependent cholesterol efflux from podocytes was found to contribute to proteinuria and glomerulosclerosis and may open new avenues for therapeutic interventions.⁵³⁰ The ability of podocytes to

take up oxidized LDL^{491,492} warrants further investigation, as LDL removal by apheresis in patients with proteinuric DKD decreases proteinuria and reduces the loss of podocytes in the urine.⁵³¹ The role of free fatty acids in the development of albuminuria in DKD also remains to be established, as free fatty acid utilization by podocytes has been described^{532,533} and may contribute to the development of podocyte-specific IR.⁴⁹⁰ Free fatty acids may also contribute to alterations in histones, linking between metabolic and epigenetic changes in DKD.⁵³⁴

Kruppel-like factor 6 (KLF6) expression was found to protect from podocyte damage in DKD. While podocyte-specific *Klf6* deficiency increases the susceptibility to STZ-induced DKD in C57BL/6 mice associated with mitochondrial injury, overexpression of KLF6 in human podocytes cultured in high glucose prevents mitochondrial injury and apoptosis. In further support of a protective effect of KLF on podocyte damage, podocyte-specific expression of KLF6 was found significantly reduced in kidney biopsies of patients with progressive DKD.⁵³⁵

Aberrant sphingolipid metabolism also contributes to podocyte injury. Inherited recessive mutations in the gene *SGPL1*, which codes for sphingosine lyase, an enzyme that degrades S1P to PE and hexadecenal at the endoplasmic reticulum, have been linked to several human diseases including congenital Charcot-Marie-Tooth neuropathy and steroid-resistant nephrotic syndrome (SRNS).⁵³⁶⁻⁵³⁸ *Sgpl1* knockout mice have increased accumulation of S1P in the kidney and recapitulate a form of glomerular pathology. The S1P/S1P receptor (S1PR) signaling pathway has gained attention due to the observation that the expression of sphingosine kinase 1 (SPHK1) was found to be increased in kidneys of patients with DKD when compared with kidneys obtained from control patients without DKD. In experimental models of DKD, STZ-induced albuminuria is exacerbated in *Sphk1*-deficient mice.⁴³⁴ Another enzyme important in regulating sphingolipid metabolism, sphingomyelin-phosphodiesterase 3b-like (SMPDL3b), was also found to play a role in DKD progression. SMPDL3b is an enzyme that was originally thought to catalyze the conversion of sphingomyelin to ceramide. However, a role for SMPDL3b in the conversion of ceramide to ceramide-1-phosphate was identified, and SMPDL3b deficiency in podocytes reduces kidney ceramide-1-phosphate, thereby contributing to IR in DKD.⁵³⁹

MESANGIAL CELLS

Mesangial expansion strongly predicts the clinical manifestations of DKD.¹²² The aberrant production of extracellular matrix in DKD has been reported and extensively studied, and the contribution of mesangial cells to glomerulosclerosis is well established.⁵⁴⁰ Genetic or pharmacologic blockade of the bradykinin 2 receptor, which is heavily expressed in mesangial cells in clinical DKD, results in profound mesangial sclerosis resembling human diabetic glomerulosclerosis without significantly affecting endothelial cells or podocytes.⁵⁴¹ The first demonstration of a functional role of miRNA in kidney disease of any type was described in DKD when miRNA192 was found to be a key mediator of mesangial cell function in DKD.⁵⁴² Enhanced JAK-STAT pathway activation in mesangial cells has been described in both clinical and experimental

DKD and has led to clinical trials of JAK2 inhibitors.^{543,544} Single-cell ribonucleic acid sequencing of glomerular cells in mouse models of DKD (streptozotocin-induced diabetic endothelial nitric oxide synthase-deficient mice) revealed reduced expression of S1PR3 in mesangial cells.⁵⁴⁵

TUBULOINTERSTITIAL CELLS

The degree of tubule-interstitial fibrosis correlates with a decline in GFR in DKD. Several urinary biomarker-based studies on how urinary NAG, KIM-1, L-FABP, and NGAL correlate with albuminuria in diabetes suggest that tubular damage starts in the early stages of DKD. In vitro and in vivo studies suggest an important role of fatty acid uptake and oxidation in the pathogenesis of CKD progression,⁵⁴⁶ and the proximal tubule seems to be a major contributor to epithelial to mesenchymal transformation.⁵⁴⁷

ADDITIONAL MECHANISMS: EPIGENETIC

Irrespective of the target cell, the role of epigenetic changes in the pathogenesis of DKD has emerged. Epigenetic modifications include cytosine methylation of DNA (DNA methylation, DNAm), histone posttranslational modifications (PTMs), and noncoding RNAs.⁵⁴⁸⁻⁵⁵⁰ A critical metabolic and epigenetic switch linking the metabolic state to chromatin remodeling is miR-93.⁵⁵¹ Epigenetic modifications may be reversible and are regulated in response to the changing environment. However, certain epigenetic changes can exhibit a memory of prior exposure to environmental cues and disease conditions, with consequent long-lasting effects even after the initial trigger has been removed. This is a finding validated by the clinical observation that the benefits of intensive insulin treatment in the EDIC study persisted long after intensive glycemic control discontinuation, suggesting the existence of metabolic memory.²⁴³ Both hemodynamic and metabolic factors may contribute to metabolic memory. Epigenetic changes occurring in the intrauterine environment can be transmitted to the offspring.⁵⁵² Different cells have distinct epigenomes.

ADDITIONAL MECHANISMS: INNATE IMMUNITY

The contribution of innate immunity by either activated local cells or infiltrating cells has also been the focus of many experimental studies.⁵⁵³ Pathogen-associated molecular patterns (PAMPs) and danger-associated molecular patterns (DAMPs) have been described as major activators of pattern recognition receptors (PRRs), such as membrane-bound Toll-like receptors and nucleotide-binding oligomerization domain (NOD)-like receptors (NLRs), which in turn cause a local inflammatory response that may be self-perpetuating irrespective of the presence or absence of the initiating stimulus. Among novel key inducers of inflammation in glomerular cells, the role of serum amyloid A (SAA) has gained much attention. SAA can stimulate NF- κ B, as well as JAK2 and PKC, leading to an autocrine amplification loop of increased endogenous production of SAA in the kidney.⁵⁵⁴ The fact that elevated baseline serum SAA can predict a composite primary outcome of death and kidney failure in a cohort of patients with type 2 diabetes and a mean baseline eGFR of 56 mL/min/1.73 m² and median UCAR of 1861 mg/g strongly suggests a potential role of causality between SSA and DKD.⁵⁵⁵ Furthermore, in a well-characterized

cohort of 74 patients with type 2 diabetes and research kidney biopsies, SAA levels correlated with AER and were higher in patients with increased GBM width. Fibrinogen and IL-6 also correlated with GBM thickening.⁵⁵⁶ On the other hand, in a cohort of patients with type 2 diabetes and early or no DKD (mean estimated GFR 128 mL/min/1.73 m² and median UCAR of 39 mg/g) higher baseline serum SAA concentration was associated with a *lower* risk of kidney failure, suggesting that early in the course of DKD, higher serum SAA concentrations may be protective.⁵⁵⁷ Identifying mechanisms that suppress early beneficial effects or promote harmful effects of SAA may enhance understanding of progressive DKD and provide new therapeutic targets.

The cyclic GMP-AMP (cGAMP) synthase (cGAS) and stimulator of interferon genes (STING) pathway is a critical player in metabolic inflammation. Analysis of experimental models of DKD in type 1 diabetes and type 2 diabetes reveal an important role for cGAS-STING activation in the glomerular but not the tubular compartment in the progression of DKD.⁵⁵⁸ In support, *Sting* deletion in podocytes reduces podocyte injury, kidney dysfunction, and inflammation via the repression of the NLRP3 inflammasome activation.⁵⁵⁹

Clinical Relevance: Pathogenesis

A large body of discovery research is now available to support the need to develop new treatment strategies that go above and beyond targeting glycemia for the treatment and prevention of DKD. The ability to develop strong systems biology approaches to understand the pathogenesis of DKD has also helped to guide discovery research toward clinically relevant targets. Rigorous experiments have helped to dissect autocrine, paracrine, and endocrine contributors to DKD and have provided evidence of a complex crosstalk between organs affected by diabetes and between specialized kidney cells. Finally, the biology of small noncoding RNAs and epigenetic contributors has shed light on the mechanism of “glucose memory” and is now offering additional therapeutic targets.

TREATMENT

Guideline-directed medical therapy of patients with type 2 diabetes and DKD has evolved dramatically in recent years owing to landmark kidney disease and cardiovascular outcome trials establishing cardio-kidney-metabolic (CKM) risk reduction with multiple new therapies when used as add-on to standard of care with ACE inhibition or ARB therapy.⁵⁶⁰⁻⁵⁶² Fig. 41.13 illustrates current consensus recommendations for the management of patients with DKD per the American Diabetes Association (ADA) and Kidney Disease: Improving Global Outcomes (KDIGO).⁵⁶³ Contemporary management of patients with type 2 diabetes and DKD employs a combination therapy approach to target multiple mechanistic drivers of kidney damage in type 2 diabetes (Fig. 41.14).⁵⁶⁴ Using Fig. 41.14 as a framework, the next sections discuss current recommended treatments for DKD, followed by a discussion of clinical trials and

investigational therapies for patients with type 1 and type 2 diabetes and CKD.

FOUNDATIONAL LIFESTYLE INTERVENTIONS

Guidelines for the management of DKD emphasize foundational lifestyle interventions, patient-centered self-management education that emphasizes CKM risk reduction, and optimized management of traditional risk factors (blood glucose, blood pressure, and lipids) for all patients with DKD (e.g., type 1 and type 2 diabetes).^{32,563} In addition to consuming a healthy diet and engaging in physical activity, encouraging and helping facilitate smoking cessation in tobacco smokers is recommended, as is achieving and maintaining weight management goals.^{32,563}

NUTRITION

Nutrition management is a keystone of the comprehensive care plan for patients with diabetes and CKD.³² Key considerations related to nutrition management include adoption of a generally healthy and balanced diet, maintaining appropriate protein intake, and avoiding excessive sodium intake. Like general recommendations for patients with diabetes, patients with DKD can benefit from individualized meal plans that stress intake of whole foods such as vegetables, fruits, whole grains, fiber, legumes, plant-based proteins, unsaturated fats, and nuts.^{30,32,563} Minimizing consumption of processed meats, refined carbohydrates, and sweetened beverages is encouraged. While dietary protein restriction has been proposed as a strategy to slow DKD progression,⁵⁶⁵ overly restrictive protein intake combined with limited carbohydrate consumption can contribute to meal plans that promote excessive weight loss.³² Therefore KDIGO recommends following the World Health Organization (WHO) recommended daily protein intake of 0.8 g/kg per day for patients with DKD not treated with dialysis.³² For dialysis patients, however, a higher daily protein intake of 1.0 to 1.2 g/kg per day is recommended. Further, to help facilitate blood pressure optimization and cardiovascular risk reduction, patients with DKD are recommended to limit sodium intake to <2 g per day.³² Overall, adherence to CKD nutritional recommendations is low.⁵⁶⁶ To address this barrier, referring patients with DKD to accredited nutrition professionals and engaging patients in a shared decision-making process to develop an individualized nutrition plan is critical to encourage healthy eating and overall promotion of healthy behavior change.

PHYSICAL ACTIVITY

Physical activity is likewise encouraged, with moderate-intensity physical activity for a cumulative duration of 150 minutes per week or a level appropriate given any cardiovascular and/or physical limitations.^{32,563} Notably, health care professionals should tailor the recommended exercise prescription for patients at increased risk for falls or other potential injuries. When individualizing a physical activity plan for patients with DKD, it should consider age, ethnic background, presence of other relevant comorbidities, and access to resources. Patients

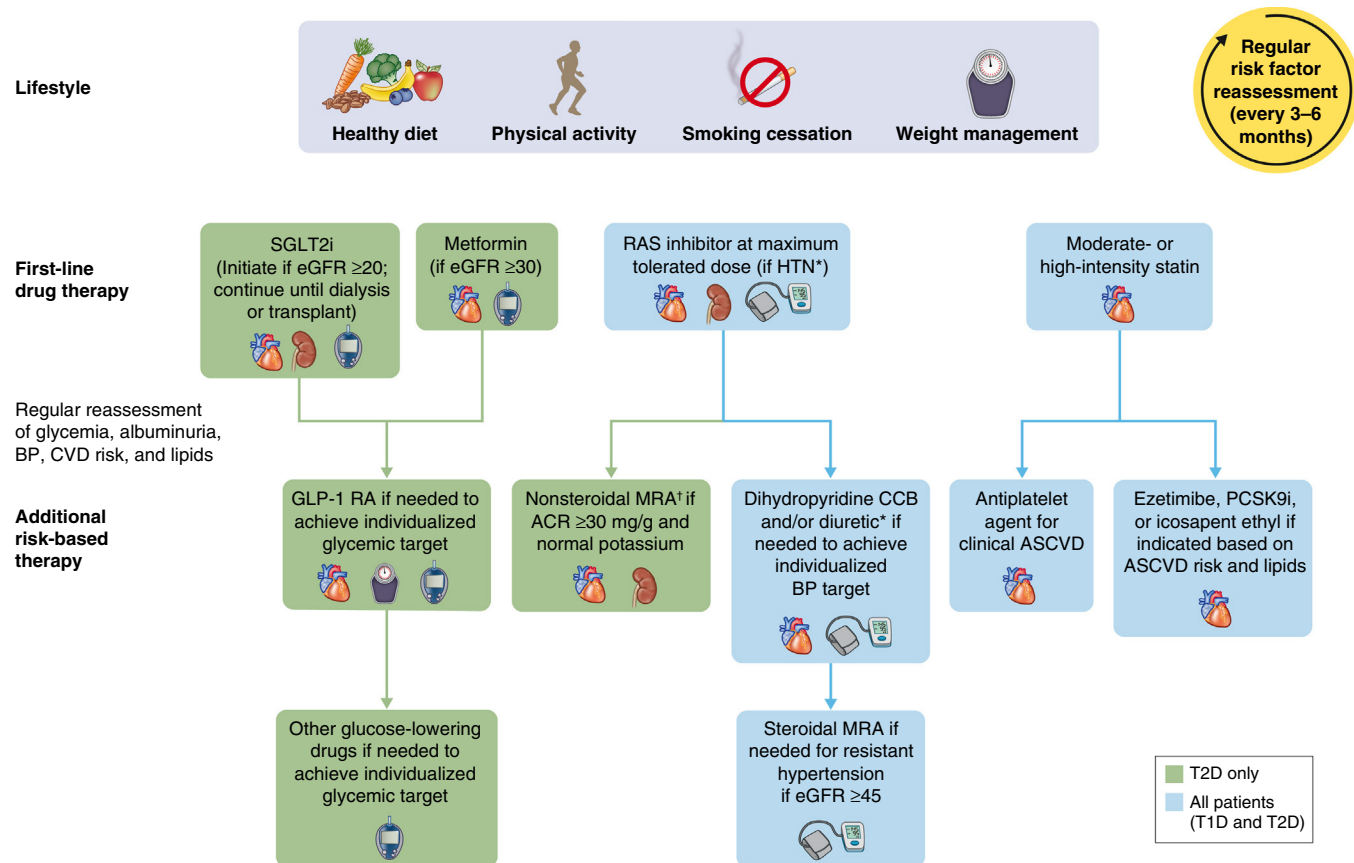


Fig. 41.13 Holistic approach for improving outcomes in patients with diabetes and chronic kidney disease (CKD).⁵⁶³ Icons presented indicate the following benefits: blood pressure (BP) cuff, BP-lowering; glucometer, glucose-lowering; heart, cardioprotection; kidney, kidney protection; scale, weight management. Estimated glomerular filtration rate (eGFR) is presented in units of mL/min/1.73 m². *Angiotensin-converting enzyme inhibitor (ACEi) or angiotensin II receptor blocker (ARB) (at maximal tolerated doses) should be first-line therapy for hypertension (HTN) when albuminuria is present. Otherwise, dihydropyridine calcium channel blocker (CCB) or diuretic can also be considered; all three classes are often needed to attain BP targets. †Finerenone is currently the only nonsteroidal mineralocorticoid receptor antagonist (ns-MRA) with proven clinical kidney and cardiovascular benefits. ACR, Albumin-to-creatinine ratio; ASCVD, atherosclerotic cardiovascular disease; CVD, cardiovascular disease; GLP-1 RA, GLP-1 receptor agonist; MRA, mineralocorticoid receptor antagonist; PCSK9i, proprotein convertase subtilisin/kexin type 9 inhibitor; RAS, renin-angiotensin system; SGLT2i, sodium-glucose cotransporter-2 inhibitor; T1D, type 1 diabetes; T2D, type 2 diabetes.

with obesity, diabetes, and CKD should be encouraged to lose weight, particularly if they have an eGFR ≥ 30 mL/min/1.73 m².³²

FIRST-LINE RECOMMENDED PHARMACOLOGIC THERAPIES

RENIN-ANGIOTENSIN SYSTEM INHIBITION

While renin-angiotensin system inhibitor therapy does not appear to prevent DKD in the absence of hypertension and albuminuria,^{567,568} ACE inhibitor or ARB therapy is protective in hypertensive patients with albuminuria.^{30,32,563,569} In patients with diabetes, eGFR < 60 mL/min/1.73 m², and albuminuria (UACR ≥ 300 mg/g), use of an ACE inhibitor or ARB reduces the risk of progression to kidney failure.⁵⁷⁰⁻⁵⁷² Treatment with a maximum tolerated dose of an ACE inhibitor or ARB is therefore recommended for all patients with DKD.^{32,563} Dual renin-angiotensin system blockade with an ACE inhibitor plus ARB therapy is not recommended as the combination was associated with increased risks for hyperkalemia and acute kidney injury.^{573,574} Table

41.5 provides a summary of recommendations from the 2022 KDIGO clinical practice guidelines regarding renin-angiotensin system inhibitor use in patients with diabetes and CKD.³²

SGLT2 INHIBITORS

Following initial signals of kidney benefit from cardiovascular outcome trials (CVOTs) with agents from the SGLT2 inhibitor class,⁵⁷⁵⁻⁵⁷⁷ three dedicated kidney outcome trials with canagliflozin, dapagliflozin, and empagliflozin, respectively, have established SGLT2 inhibitors as a first-line standard of care for people with type 2 diabetes and CKD (Table 41.6).⁵⁷⁸⁻⁵⁸⁰ Importantly, all three trials tested SGLT2 inhibitor treatment as add-on to background renin-angiotensin system inhibitor therapy. The first dedicated kidney outcome trial published was the Canagliflozin and Renal Events in Diabetes with Established Nephropathy Clinical Evaluation (CREDENCE) trial.⁵⁷⁸ CREDENCE by design enrolled participants with type 2 diabetes and macroalbuminuric CKD. A 30% relative risk reduction for its primary kidney disease composite

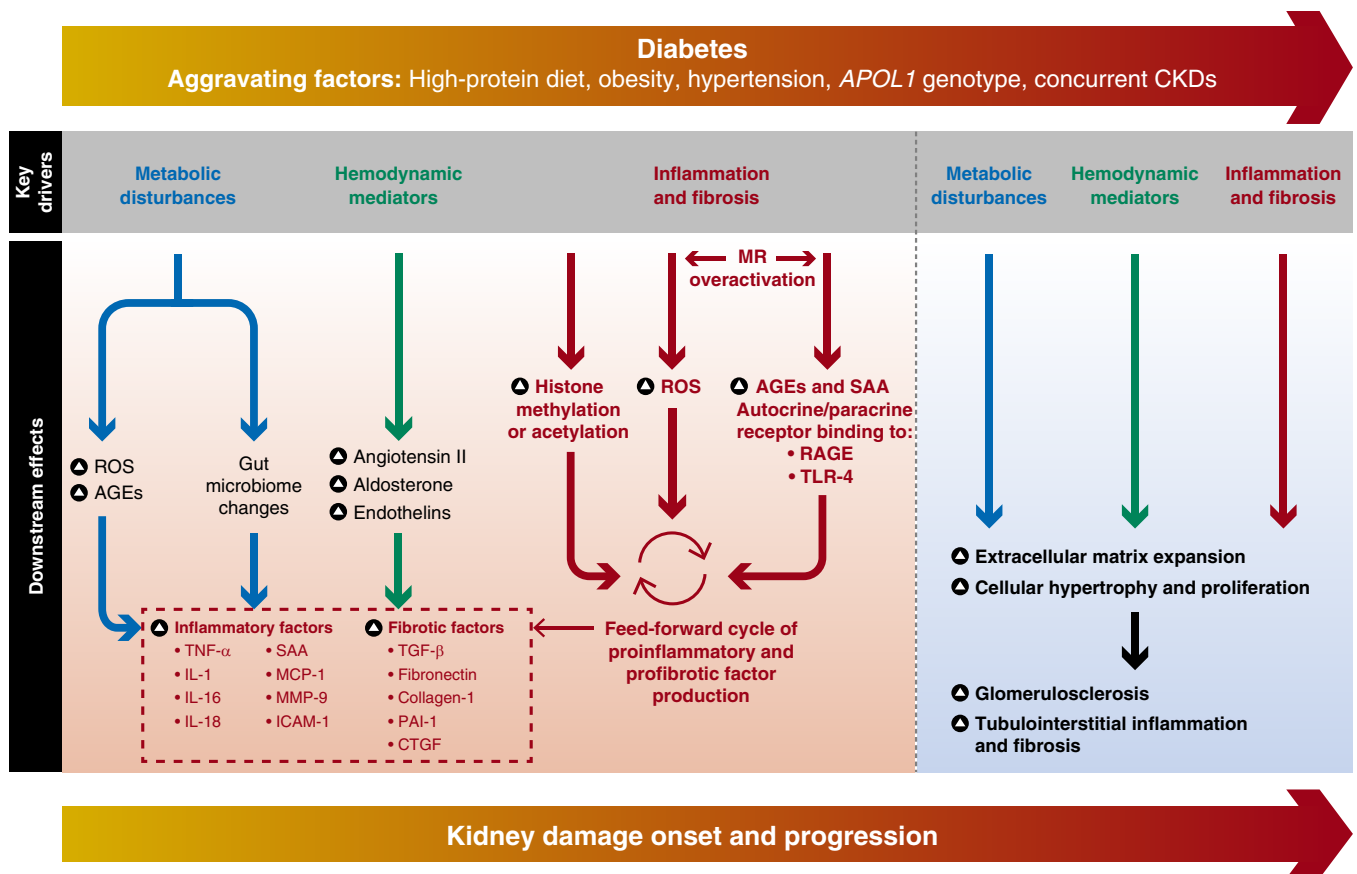


Fig. 41.14 Diagram showing the interrelation of mechanistic drivers in early through advanced stages of kidney damage and disease progression in diabetes.⁵⁶⁴ AGE, Advanced glycation end product; CKD, chronic kidney disease; CTGF, connective tissue growth factor; DKD, diabetic kidney disease; ICAM-1, intracellular adhesion molecule 1; IL, interleukin; MCP-1, monocyte chemoattractant protein-1; MMP-9, matrix metalloproteinase 9; MR, mineralocorticoid receptor; PAI-1, plasminogen activator inhibitor; RAGE, receptor for advanced glycation end product; ROS, reactive oxygen species; SAA, serum amyloid A; TGF- β , transforming growth factor- β ; TLR-4, Toll-like preceptor-4; TNF- α , tumor necrosis factor- α .

outcome was observed (Table 41.6).⁵⁷⁸ The Dapagliflozin and Prevention of Adverse Outcomes in Chronic Kidney Disease (DAPA-CKD) and the Study of Heart and Kidney Protection With Empagliflozin (EMPA-KIDNEY) trials subsequently demonstrated kidney protection in populations with and without type 2 diabetes, participants with baseline eGFRs down to 20 mL/min/1.73 m², and with varying degrees of albuminuria at baseline.^{579,580} Subgroup analyses from the EMPA-KIDNEY trial found no heterogeneity of effect based on background diabetes or baseline eGFR,⁵⁸⁰ findings supported by a meta-analysis that included multiple large SGLT2 inhibitor outcome trials.⁵⁸¹

Research to understand the mechanisms responsible for kidney benefits with SGLT2 inhibitors is ongoing.⁵⁸² A principal mechanism of action of SGLT2 inhibitors is blockade of glucose reabsorption in the proximal tubule, which is upregulated in diabetes (Fig. 41.4).⁵⁸³ Inhibition of SGLT2, in turn, leads to glucosuria and increased delivery of sodium chloride to the macula densa.⁵⁶⁴ Increased solute reabsorption increases ATP utilization and generation of adenosine that acts in a paracrine manner to increase afferent vasoconstriction, and possibly efferent vasodilation in type 2 diabetes, thus

ameliorating glomerular hypertension (Fig. 41.15).^{80,584} Other proposed mechanisms of benefit mediated by SGLT inhibition are mitigation of oxidative stress, reduction of inflammation, reduction of intracellular glucose levels, and increasing erythropoietin production in a variety of kidney, heart, and endothelial cells.^{582,585-590} As agents initially approved to lower blood glucose, SGLT2 inhibitors may also convey kidney and heart protection through improved glycemic control, blood pressure reduction, weight loss, and a shift in energy substrate utilization to lipids.^{583,591}

Risk mitigation strategies for SGLT2 inhibitor side effects include hygiene counseling to avoid genital mycotic infections, provision of sick day rules and insulin administration guidelines to reduce risk of SGLT2 inhibitor-associated “normoglycemic” ketoacidosis, and proactive reduction of diuretics as needed for patients at risk for hypovolemia or hypotension.^{32,563,592} SGLT2 inhibitor initiation is associated with a reversible decline in eGFR of 3 to 5 mL/min/1.73 m². After this initial “eGFR dip,” kidney function typically stabilizes during ongoing SGLT2 inhibitor therapy.^{593,594} Because SGLT2 inhibitor therapy can lower glycemia in patients with preserved kidney function, generally eGFR >45 mL/min/1.73 m², consideration of

the need for adjusting background insulin secretagogue or insulin therapies may be appropriate to avoid hypoglycemia (Table 41.7).^{32,563}

Table 41.5 KDIGO Practice Points for use of Renin-Angiotensin System Blockers in Patients with Diabetic Kidney Disease³²

- ACE inhibitor or ARB therapy may be considered in patients with diabetes, albuminuria, and normal blood pressure.
- After starting or increasing the dose of an ACE inhibitor or ARB, monitor for changes in blood pressure, serum creatinine, and serum potassium within 2-4 weeks.
- ACE inhibitor/ARB therapy should be continued unless serum creatinine increases by more than 30% within 4 weeks of drug initiation or an increase in dose.
- Women receiving an ACE inhibitor or ARB should be advised about appropriate contraception and should be advised to discontinue therapy if they are considering pregnancy or become pregnant.
- Hyperkalemia associated with ACE inhibitor/ARB use can often be managed by interventions to reduce serum potassium levels without decreasing the dose or stopping the ACE inhibitor or ARB immediately.
- Reduce the dose or discontinue ACE inhibitor or ARB therapy with either symptomatic hypotension or uncontrolled hyperkalemia despite alternative medical treatment approaches, or to reduce uremic symptoms in the setting of kidney failure.
- Do not use dual renin-angiotensin system blockade.

ACE, Angiotensin-converting enzyme; ARB, angiotensin II receptor blocker; KDIGO, Kidney Disease Improving Global Outcomes.

ADDITIONAL RISK-BASED THERAPIES

GLP-1 AND GLP-1/GIP RECEPTOR AGONISTS

GLP-1 and glucose-dependent insulinotropic polypeptide (GIP) are incretin hormones secreted by intestinal enteroendocrine cells in response to glucose ingestion.⁵⁹⁵⁻⁵⁹⁷ Incretin hormones also play important roles in stimulating insulin secretion in the fed state in addition to regulating food intake and appetite.^{595,597,598} In addition to the established atherosclerotic cardiovascular disease benefits of agents within the GLP-1 receptor agonist class,⁵⁹⁹ evidence continues to build substantiating GLP-1 receptor agonist benefit on DKD.⁶⁰⁰ Several GLP-1 receptor agonist CVOTs reported benefits on secondary kidney disease outcomes.⁶⁰¹⁻⁶⁰³ In addition, a glycemic control trial in participants with type 2 diabetes and moderate-to-severe CKD (inclusive of patients with eGFRs down to 15 mL/min/1.73 m²) found a reduced rate of eGFR decline with dulaglutide treatment when compared with insulin glargine.⁶⁰⁴ Importantly, this trial also established glycemic lowering efficacy and safety in patients with moderate-to-severe CKD. Recent pooled analyses that included participants from liraglutide and semaglutide CVOTs further reported albuminuria reductions and slowed eGFR decline compared with placebo, with greatest benefit observed in those with baseline eGFR <60 mL/min/1.73 m².^{605,606} The trial “Effect of Semaglutide Versus Placebo on the Progression of Renal Impairment in Subjects With Type 2 Diabetes and Chronic Kidney Disease” (FLOW) provided the first primary kidney outcome data with a GLP-1 receptor agonist in the setting of type 2 diabetes and CKD. After median follow-up of 3.4 years treatment with subcutaneous semaglutide 1.0 mg weekly reduced the risk of the primary composite outcome (onset of kidney failure, at least a 50% reduction in the eGFR from baseline, or death from kidney-related or cardiovascular causes) by 24%. Both the kidney and CV components of the composite endpoint contributed to the reduction in risk,

Table 41.6 SGLT2 Inhibitor Kidney Outcome Trials Summary⁵⁷⁸⁻⁵⁸⁰

Study	CREDENCE (N = 4401)	DAPA-CKD (N = 4304)	EMPA-KIDNEY (N = 6609)
Agent	Canagliflozin	Dapagliflozin	Empagliflozin
Median follow-up	2.6 years	2.4 years	2.0 years
Diabetes status	T2D	T2D or non-T2D CKD	T2D or non-T2D CKD
Kidney-related enrollment criteria	• eGFR ≥30 to <90 mL/min/1.73 m² • UACR: ≥300 to <5000 mg/g	• eGFR ≥25 to <75 mL/min/1.73 m² • UACR: ≥200 to 5000 mg/g	• eGFR ≥20 to <45 mL/min/1.73 m² OR • eGFR ≥45 to <90 mL/min/1.73 m² with UACR ≥200 mg/g
Mean baseline eGFR	56 mL/min/1.73 m ²	43 mL/min/1.73 m ²	37.5 mL/min/1.73 m ²
Median baseline UACR	927 mg/g	949 mg/g	412 mg/g
Kidney outcome(s)	Primary Outcome • ESKD (dialysis, transplantation, or sustained eGFR <15 mL/min/1.73m²), doubling of SCr, or death from renal causes: HR: 0.70 (0.59-0.82)	Primary Outcome • ≥50% decrease in eGFR, ESKD, or death from renal or cardiovascular causes: HR: 0.61 (0.51-0.72)	Primary Outcome* • ≥40% decrease in eGFR, ESKD, sustained decline in eGFR to <10 mL/min/1.73m² or cardiovascular death: HR: 0.72 (0.64-0.82)

eGFR, Estimated glomerular filtration rate; ESKD, end-stage kidney disease; HR, hazard ratio; SCr, serum creatinine; T2D, type 2 diabetes; UACR, urinary albumin-to-creatinine ratio.

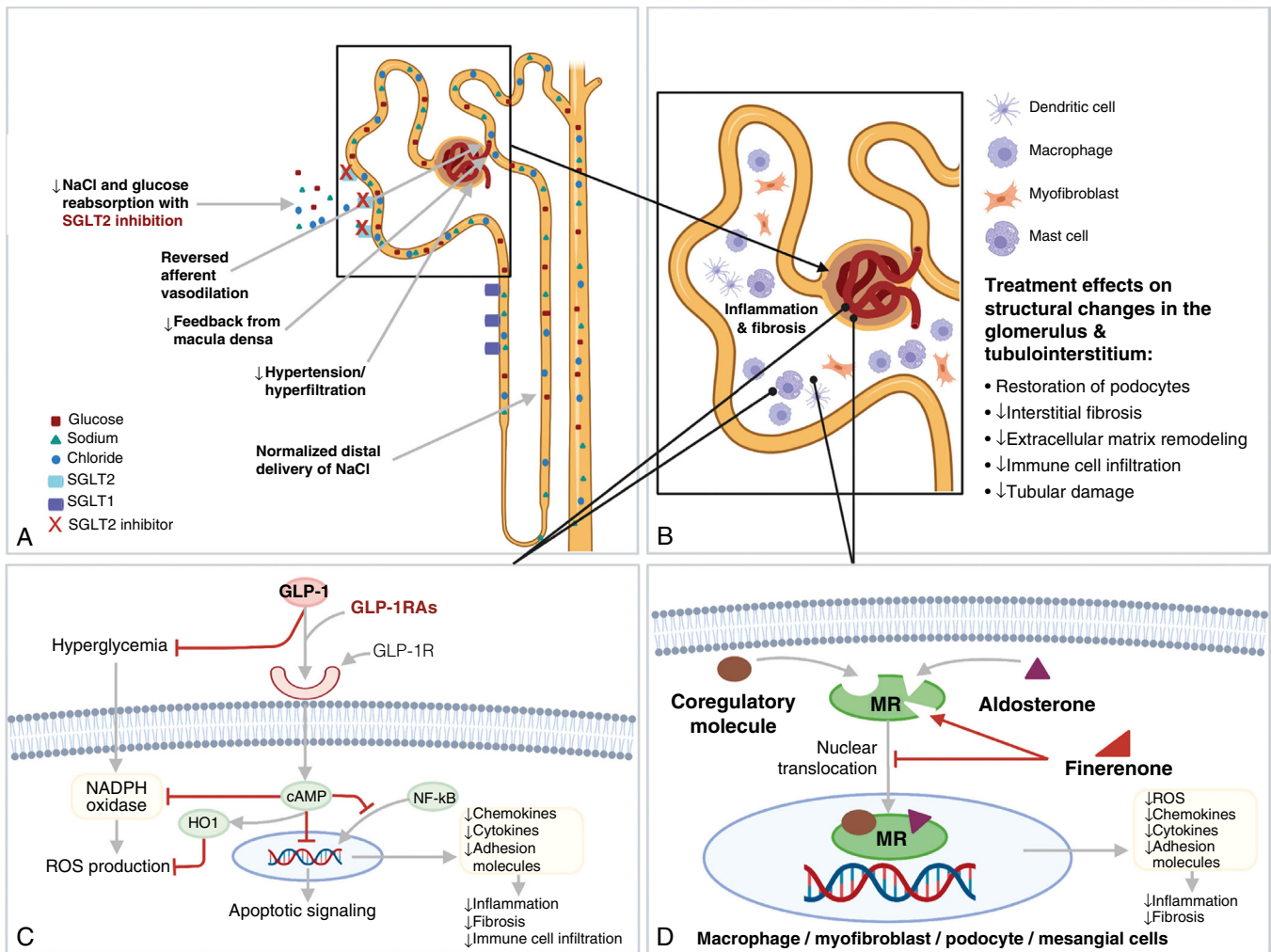


Fig. 41.15 Mechanisms of kidney protection with SGLT2 inhibitors, GLP-1RAs, and a nonsteroidal mineralocorticoid receptor antagonist (MRA).⁵⁸⁴ (A) Hemodynamic changes in the diabetic kidney are reversed with SGLT2 inhibition. In diabetes, the resorptive capacity for glucose in the proximal tubule is increased via upregulation of SGLT2 and SGLT1. As a result of enhanced glucose and sodium chloride uptake in the proximal tubule, solute delivery to the macula densa cells of the juxtaglomerular apparatus is diminished, resulting in altered tubuloglomerular feedback. Adenosine release is subsequently decreased resulting in vasodilation of afferent arteriola, glomerular hyperfiltration, and hypertension. SGLT2 inhibition decreases glucose/sodium reabsorption, thus increasing solute delivery to the distal tubule. These effects help restore tubuloglomerular feedback with a resulting increase in production of adenosine leading to vasoconstriction of afferent arteriola, with improvement of glomerular hyperfiltration and hypertension. (B) Structural changes observed in patients with diabetes and chronic kidney disease include glomerular hypertrophy, thickening of the glomerular basement membrane, podocyte detachment and foot-process effacement, expansion of glomerular mesangial cell matrix, immune cell infiltration, and interstitial fibrosis. Treatment with SGLT2 inhibitors, GLP-1RAs, and ns-MRA helps restore podocytes and decreases extracellular mesangial matrix remodeling, immune cell infiltration, tubular damage, and interstitial inflammation and fibrosis. (C) The proposed effects of GLP-1RAs in kidney are predominantly mediated through activation of the GLP-1 receptor (GLP1R). Beneficial effects are principally related to suppression of inflammation and oxidative stress, reduced immune cell infiltration, and reduced fibrosis. Activation of the GLP1R reduces production of reactive oxygen species (ROS) via haem-oxygenase 1 (HO1) and reduces production of proinflammatory chemokines, cytokines, adhesion molecules, and profibrotic factors via inhibition of nuclear factor-κB (NF-κB) binding. ROS production is also reduced through a non-receptor-mediated reduction in nicotinamide adenine dinucleotide phosphate (NADPH) oxidase. (D) Overactivation of the MR in diabetes has further been implicated in the promotion of inflammation and fibrosis. Antagonism of the mineralocorticoid receptor with steroidal (e.g., spironolactone) and nonsteroidal (finerenone) MRAs suppresses expression of proinflammatory and profibrotic genes in macrophages, myofibroblasts, podocytes, and mesangial cells. cAMP, Cyclic adenosine monophosphate; GLP1, glucagon-like peptide 1; GLP1R, glucagon-like peptide 1 receptor; GLP-1RA, glucagon-like peptide 1 receptor agonist; HO1, haem-oxygenase 1; MR, mineralocorticoid receptor; NaCl, sodium chloride; NADPH, nicotinamide adenine dinucleotide phosphate; NF-κB, nuclear factor-κB; ROS, reactive oxygen species; ns-MRA, nonsteroidal mineralocorticoid receptor antagonist; SGLT-1, sodium-glucose cotransporter 1; SGLT-2, sodium-glucose cotransporter 2.

with a 21% risk reduction observed for the kidney-specific components of the primary outcome and a 29% reduction in death from cardiovascular causes. Importantly, the risk of death from any cause was 20% lower in the semaglutide group compared with placebo.⁶⁰⁷ Evidence additionally

continues to build regarding the impact of the dual GLP-1/GIP receptor agonist, tirzepatide, on kidney and CVD outcomes. One report highlighted exploratory benefits of tirzepatide treatment on both albuminuria and eGFR decline.⁶⁰⁸

Table 41.7 Key Monitoring and Risk Mitigation Strategies for Agents Recommended in Patients with type 2 Diabetes and Chronic Kidney Disease^{32,563}

Medication class	Consideration	Monitoring and/or risk mitigation strategies
SGLT2 inhibitors	• Volume depletion • Diabetic ketoacidosis (DKA) • Genital mycotic infections • Hypoglycemia	• Proactive dose reduction of diuretics in patients at high risk for hypovolemia • Hold SGLT2 inhibitors during illness • Educate about signs/symptoms to facilitate early recognition • Monitor blood or urine ketones for very high risk • Institute a sick day protocol • Maintain at least low-dose insulin in insulin-requiring individuals • Counsel on genital hygiene (e.g., regular bathing and wearing clean undergarments) • Adjust background glucose-lowering agents (e.g., insulin and/or sulfonylureas) as appropriate
Metformin	• Metformin-associated lactic acidosis • B₁₂ malabsorption	• Monitor eGFR with increasing frequency as eGFR falls <60 mL/min/1.73 m² • Adjust metformin dose as appropriate per eGFR • Consider dose reduction in the presence of conditions that predispose patients to hypoperfusion and hypoxemia if eGFR 45–59 mL/min/1.73 m² • Institute a sick day protocol • Monitor patients for vitamin B₁₂ deficiency when treated with metformin for >4 years
GLP1 receptor agonists	• Nausea/vomiting/diarrhea • Hypoglycemia	• Start at lowest recommended dose and titrate slowly • Educate on potential GI intolerance and importance of communicating severe symptoms to care team • Adjust background glucose-lowering agents (e.g., insulin and/or sulfonylureas) as appropriate
Nonsteroidal MRAs	• Hyperkalemia	• Monitor serum potassium prior to initiation and during treatment

eGFR, Estimated glomerular filtration rate; GI, gastrointestinal; GLP1, glucagon-like peptide-1; SGLT2, sodium–glucose cotransporter-2.

Metabolic benefits of GLP-1RAs are well established, with preserved glucose-lowering effects in advanced CKD, reduction of blood pressure by ~3 to 4 mm Hg, and a mean weight reduction of approximately 3 kg noted in clinical trials.^{609–611} While metabolic benefits are preserved in CKD, kidney, and heart, protective effects are not fully explained by reductions in glycemia, blood pressure, and weight. For instance, the relative impact of HbA1c, systolic blood pressure, and body weight change on kidney outcomes was assessed in a mediation analysis of completed CVOTs with liraglutide and semaglutide.⁶¹² The analysis reported that lower glycemia and reduced blood pressure only moderately mediated (10%–25%) development of macroalbuminuria, doubling of serum creatinine, decline in eGFR to <45 mL/min/1.73 m², and progression to kidney failure, thus pointing to direct protective effects of GLP-1RAs in the kidney.⁶¹² Indeed, mechanistic studies suggest that GLP-1 receptor agonists suppress oxidative stress, inhibit activation and infiltration of inflammatory cells into the heart and kidney, and reduce activation of proinflammatory cytokines and profibrotic factors (Fig. 41.15).^{584,613–618}

Table 41.7 highlights key risk mitigation strategies to consider with use of GLP-1 receptor agonists.^{32,563} Gastrointestinal side effects are the most common dose-limiting side effects with GLP-1 receptor agonists and tirzepatide. Starting at the lowest dose and titrating judiciously to maximize tolerability is recommended. Counseling patients on the importance of eating smaller

portions, eating slowly, stopping eating once full, and avoiding high-fat and/or spicy foods can help minimize gastrointestinal intolerance.⁶¹⁹ Given the preserved glucose-lowering effect of incretin therapies in the setting of CKD, adjusting background insulin secretagogue or insulin therapies should be considered to prevent hypoglycemia.

NONSTEROIDAL MINERALOCORTICOID RECEPTOR ANTAGONISTS (NS-MRAs)

Overactivation of the MR is associated with inflammation and fibrosis. Therefore the MR has emerged as a therapeutic target to improve kidney and heart outcomes in patients with type 2 diabetes and CKD.^{620,621} MRs are located in distal tubules, the collecting duct, podocytes, fibroblasts, and mesangial cells.⁶²² Finerenone is the only nonsteroidal MRA (ns-MRA) currently approved for use in the United States. It selectively binds to the MR as a “bulky” antagonist.⁶²⁰ Once bound, the finerenone-MR complex transits to the nucleus and downregulates proinflammatory and profibrotic gene transcription (Fig. 41.15).⁵⁸⁴ This action in turn inhibits transcriptional cofactor recruitment involved in hypertrophic, proinflammatory, and profibrotic gene regulation.⁶²³ The kidney and heart benefits of finerenone were established through two primary outcome trials: the Effect of Finerenone on Chronic Kidney Disease Outcomes in Type 2 Diabetes (FIDELIO-DKD) trial and the Finerenone in Reducing Cardiovascular Mortality and Morbidity in Diabetic Kidney Disease (FIGARO-DKD)

Table 41.8 Summary of Finerenone Outcome Trials^{624,625}

Trial	FIDELIO-DKD (n = 5734)	FIGARO-DKD (n = 7437)
Treatment	Finerenone vs. placebo	Finerenone vs. placebo
Mean participant age (years)	66	64
Key inclusion criteria	• T2D • eGFR 25 to <60 mL/min/1.73 m² and UACR 30 to <300 mg/g, or • eGFR 25 to <75 mL/min/1.73 m² and UACR 300 to 5000 mg/g • Treated with RAS inhibitor at maximum tolerated dose	• T2D • eGFR 25 to 90 mL/min/1.73 m² and UACR 30 to <300 mg/g, or • eGFR >60 mL/min/1.73 m² and UACR 300 to 5000 mg/g • Treated with RAS inhibitor at maximum tolerated dose
Mean baseline A1C (%)	7.7	7.7
Median follow-up (years)	2.6	3.4
Primary outcome		
HR (95% CI)	Kidney failure, ≥40% decline in eGFR, or renal death 0.82 (0.73-0.93)	CV death, nonfatal MI, nonfatal stroke, or hospitalization for HF 0.87 (0.76-0.98)
Key secondary outcomes		
Key secondary composite; HR (95% CI)	CV death, nonfatal MI, nonfatal stroke, or hospitalization for HF 0.86 (0.75-0.99)	Kidney failure, ≥40% decline in eGFR, or renal death 0.87 (0.76-1.01)

A1C, Glycated hemoglobin A1c; CI, confidence interval; CV, cardiovascular; eGFR, estimated glomerular filtration rate; HF, heart failure; HR, hazard ratio; MI, myocardial infarction; RAS, renin-angiotensin system; T2D, type 2 diabetes mellitus; UACR, urinary albumin-to-creatinine ratio.

trial (Table 41.8).^{624,625} FIDELIO-DKD and FIGARO-DKD reported relative risk reductions of 18% and 13% for their primary kidney and cardiovascular composite outcomes, respectively. Similar to kidney disease outcome studies with SGLT2 inhibitors, the cardiovascular and kidney benefits of finerenone were observed when added to optimized background renin-angiotensin system inhibitor therapy. A pooled analysis of data from both trials reported a benefit of finerenone treatment on composite kidney and cardiovascular outcomes across a range of baseline eGFR and albuminuria levels,⁶²⁶ irrespective of prevalent atherosclerotic cardiovascular disease.⁶²⁷ While finerenone may have a lower risk for hyperkalemia compared with steroidal MRAs, potassium monitoring and management remain essential to prevent and identify hyperkalemia. Serum potassium should be monitored before drug initiation and periodically during treatment to inform dose titration (Table 41.7).^{32,563}

MANAGEMENT OF MODIFIABLE RISK FACTORS IN PATIENTS WITH DIABETIC KIDNEY DISEASE

GLYCEMIC MANAGEMENT

Optimization of glucose control to reduce risks of microvascular complications is considered a cornerstone of glycemic management,^{32,563,569} with “intensive

glucose control” delaying the onset and progression of albuminuria and eGFR decline in both type 1^{628,629} and type 2 diabetes.^{66,630-633} While the importance of glycemic management is clear, impaired kidney function impacts the risks and benefits association with use of glucose-lowering therapies. Many glucose-lowering medications are cleared by the kidney and require dose adjustment (Fig. 41.16).⁵⁶³ The impact of reduced kidney function on hypoglycemia and/or increased risks for other adverse drug events must also be considered.^{29,634}

Metformin

ADA and KDIGO additionally include metformin as an optional first-line therapy if needed for glycemic control in those with eGFR ≥30 mL/min/1.73 m².⁵⁶³ Metformin is included on the basis of its long track record of safety, efficacy, accessibility, and affordability. While risk for metformin-associated lactic acidosis is low,⁶³⁵ metformin should be dose adjusted on the basis of eGFR and patients should be counseled about sick day rules and holding metformin during acute illness to prevent metformin accumulation and risk of lactic acidosis (Fig. 41.17).^{32,563} Due to the association with dose- and treatment duration-related B₁₂ deficiency with metformin use,⁶³⁶ B₁₂ monitoring is also recommended in patients treated with metformin for >4 years (Table 41.7).³²

	Stage 3b (eGFR 30–44 mL/min/1.73 m ²)	Stage 4 (eGFR 15–29 mL/min/1.73 m ²)	Stage 5 (eGFR <15 mL/min/1.73 m ²)
Metformin	Reduce dose to 1000 mg/day	Contraindicated	
Insulin	Initiate and titrate conservatively to avoid hypoglycemia		
SGLT2 inhibitors*			
Canagliflozin	Maximum 100 mg daily	Initiation not recommended; may continue 100 mg daily if tolerated for kidney and CV benefit until dialysis	
Dapagliflozin	10 mg daily [†]	Initiation not recommended with eGFR <25 mL/min/1.73 m ² ; may continue if tolerated for kidney and CV benefit until dialysis	
Empagliflozin	10 mg daily [‡]	Initiation not recommended with eGFR <20 mL/min/1.73 m ² ; may continue if tolerated for kidney and CV benefit until dialysis	
Ertugliflozin	Use not recommended with eGFR <45 mL/min/1.73 m ²		
GLP-1 receptor agonists [§]			
Exenatide	Caution initiating or increasing dose; avoid once-weekly formulation	Use not recommended	
Dulaglutide	No dose adjustment required		
Liraglutide	No dose adjustment required		
Lixisenatide	No dose adjustment required		Use not recommended
Semaglutide	No dose adjustment required		
DPP-4 inhibitors			
Alogliptin	Maximum 12.5 mg daily	Maximum 6.25 mg daily	
Linagliptin	No dose adjustment required		
Saxagliptin	Maximum 2.5 mg daily		
Sitagliptin	Maximum 50 mg daily	Maximum 25 mg once daily	
Sulfonylureas (2nd generation)			
Glimepiride	Initiate conservatively at 1 mg daily and titrate slowly to avoid hypoglycemia		
Glipizide	Initiate conservatively (e.g., 2.5 mg once daily) and titrate slowly to avoid hypoglycemia		
Glyburide	Use not recommended		
Thiazolidinediones			
Pioglitazone	No dose adjustment required		
α-Glucosidase inhibitors			
Acarbose	No dose adjustment required	Use not recommended	
Miglitol	No dose adjustment required	Use not recommended	

Fig. 41.16 Manufacturer-recommended dose adjustments for glucose-lowering agents with eGFR <45 mL/min/1.73 m².⁵⁶³ *Glucose-lowering efficacy is reduced with SGLT2i as eGFR declines, but kidney and cardiovascular benefits are preserved. [†]Dapagliflozin is approved for use at 10 mg once daily with an eGFR of 25 to <45 mL/min/1.73 m². [‡]Initiation not recommended with eGFR <30 mL/min/1.73 m² for glycemic control or <20 mL/min/1.73 m² for CKD or HF. [§]Dulaglutide, liraglutide, and injectable semaglutide have demonstrated evidence of cardiovascular benefit in large cardiovascular outcome trials. CV, Cardiovascular; DPP-4, dipeptidyl peptidase 4; GFR, glomerular filtration rate; GLP-1, glucagon-like peptide-1; SGLT2, sodium-glucose cotransporter 2.

Glycemic targets

The 2022 KDOQI recommendations for diabetes management in CKD suggest an HbA1c target ranging from <6.5% to <8.0% in patients not treated with dialysis.³² It is important to consider HbA1c that has important limitations in the setting of CKD, thus individualization based on patient- and medication-specific factors is recommended. Targeting HbA1c <6.5% may be appropriate in individuals where prevention of additional complications is a major

consideration, while a more liberal target (e.g., HbA1c <8.0%) may be more appropriate in those with a high comorbidity burden and/or a high risk of hypoglycemia or hypoglycemia unawareness.³² This approach of individualized targets is consistent with the general approach recommended by the ADA for people with diabetes.⁵⁶⁹ Table 41.9 provides a summary of factors that may guide HbA1c target selection in patients with DKD.^{32,569} For patients where a lower HbA1c target is desirable, and adequate resources are available,

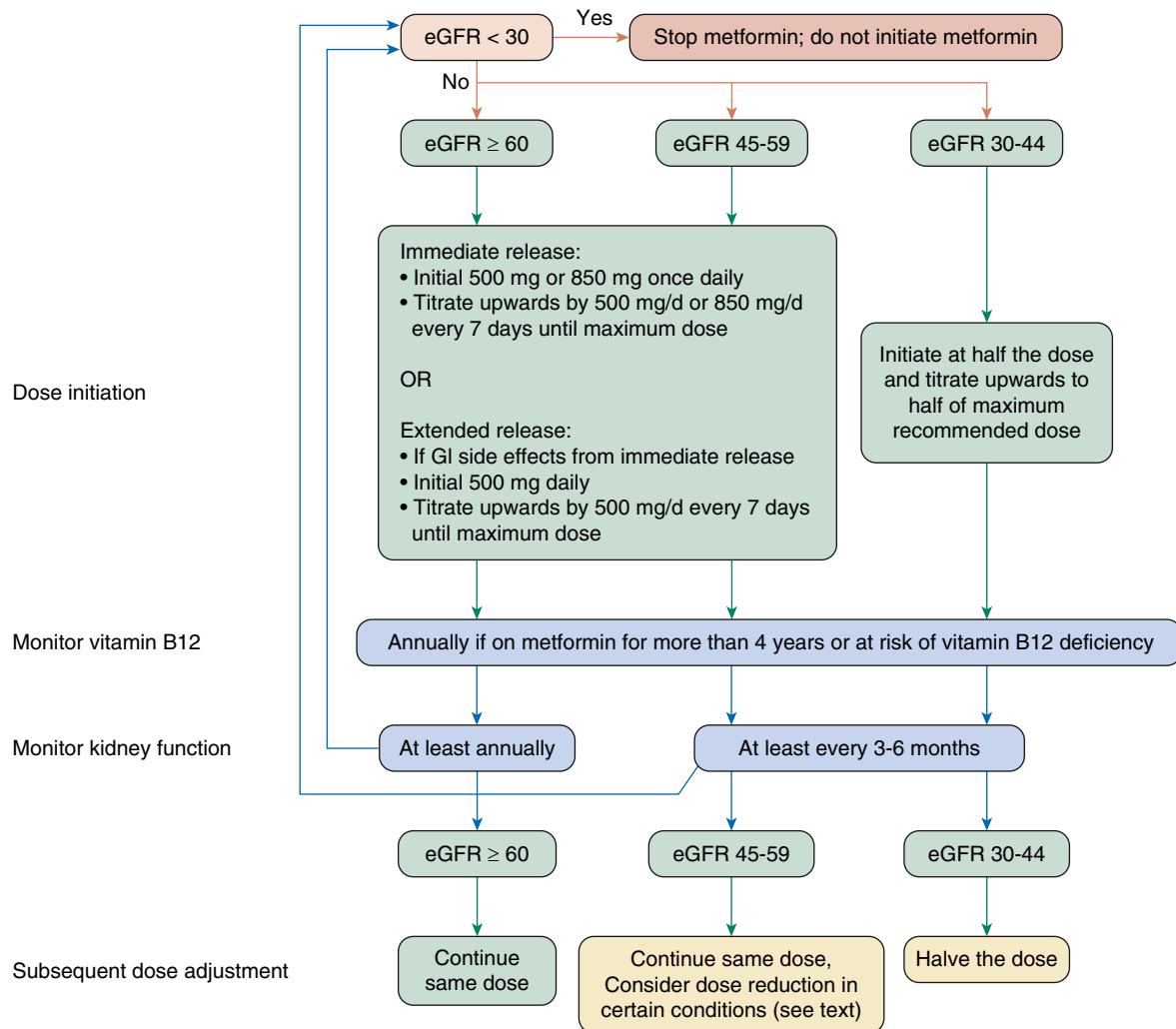


Fig. 41.17 Suggested approach in dosing metformin based on the level of kidney function.³²

Table 41.9 Factors that May Guide Decisions on Individualized A1C Targets in Diabetic Kidney Disease^{30,32,563,569}

Factor/Consideration	Supports less stringent A1C target (e.g., <8.0%)	Supports more stringent A1C target (e.g., <6.5% or 7.0%)
CKD severity	Advanced-stage CKD (e.g., CKD G5)	Early-stage CKD (e.g., CKD G1)
Macrovascular complications	Present/Severe	Absent/Minor
Comorbidity burden	High	Low
Life expectancy	Short	Long
Hypoglycemia risk	High	Low
Hypoglycemia awareness	Impaired	Present
Resources/Support	Scarce	Available
Patient motivation/preference for treatment intensity	Low motivation	Highly motivated

Adapted from El Sayed NA, Aleppo G, Aroda VR, et al. American Diabetes Association. Standards of Care in Diabetes—2023. *Diabetes Care*. 2023;46(suppl 1):S1–S291 and Kidney Disease: Improving Global Outcomes Diabetes Work Group. KDIGO 2022 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease. *Kidney Int*. 2022;102:S1–S127.

continuous glucose monitoring (CGM) or fingerstick blood glucose monitoring (BGM) can assist in efforts to intensify therapy while minimizing hypoglycemia risk. For patients where HbA1c may not be a reliable indicator of overall glycemic control and use of CGM is an option, CGM metrics such as time in range and time spent in hypoglycemia (defined as time spent <70 mg/dL) are helpful metrics to guide daily care and treatment modification.^{32,637}

Glycemic monitoring

HbA1c is the recommended metric for routine monitoring of long-term glycemic control in patients with DKD.^{32,569} While HbA1c monitoring is recommended in the setting of DKD, it is important to consider its limitations in this patient population. HbA1c reflects glycation of red blood cells over the lifespan of the erythrocyte. The HbA1c value can be decreased by factors that reduce the erythrocyte lifespan such as anemia, following blood transfusion, and during use of erythrocyte-stimulating agents or iron replacement.^{638,639} Accordingly, the precision and accuracy of HbA1c as a metric of glycemic control is unreliable in advanced CKD, particularly in patients undergoing dialysis. Nonetheless, HbA1c remains the biomarker of choice because alternative methods do not have appreciable advantages over HbA1c testing. Both CGM and BGM are additionally recommended to inform daily treatment decisions.^{32,637} CGM and BGM measurements are not known to be biased by CKD or its treatments and can also be effective tools to validate a correlate of HbA1c, the glucose management indicator (GMI).³² In patients where the reliability of the HbA1c is questionable (e.g., chronic hemodialysis), CGM can be used to generate a GMI to inform treatment decisions, which is a calculated measure of average glucose control generated from CGM data—a particularly useful measure in the setting of advanced CKD.⁶³⁷

BLOOD PRESSURE MANAGEMENT

Blood pressure management is recommended by all major guideline-forming organizations for prevention of CKD progression, atherosclerotic cardiovascular disease, and heart failure risks.^{30,32,563,640} The KDIGO 2021 Clinical Practice Guideline for the Management of Blood Pressure in Chronic Kidney Disease recommends a systolic blood pressure target of <120 mm Hg if it can be achieved without undue risk or treatment burden.³² KDIGO and ADA emphasize the importance of individualized targets based on careful consideration of risks and potential benefits. A renin-angiotensin system inhibitor titrated to the maximum tolerated dose indicated is recommended for people with diabetes, CKD, and albuminuria.^{30,32,563,640} Additional antihypertensive therapies can be added to achieve individualized targets, with dihydropyridine calcium channel blockers and thiazide diuretics recommended by the ADA as other preferred antihypertensive medication classes.⁶⁴¹ Please refer to Chapter 46 for a more detailed discussion of hypertension management in patients with kidney disease.

LIPID-LOWERING THERAPY

There is a strong relationship between DKD and cardiovascular disease, with much of the mortality seen in DKD attributable to cardiovascular disease.²⁹ Indeed,

Table 41.10 Moderate- and High-Intensity Statin Therapy⁶⁴¹

Moderate intensity (lowers LDL by 30%-49%)	High intensity (lowers LDL by ≥50%)
• Atorvastatin 10-20 mg • Rosuvastatin 5-10 mg • Simvastatin 20-40 mg • Pravastatin 40-80 mg • Lovastatin 40 mg • Fluvastatin XL 80 mg • Pitavastatin 1-4 mg	• Atorvastatin 40-80 mg • Rosuvastatin 20-40 mg

All dosages are once-daily dosing.
XL, Extended release.

albuminuria and eGFR are independently and additively associated with an increased risk for cardiovascular disease events and mortality, as well as all-cause death.³⁴³ Given the relationship between DKD and cardiovascular disease, treatment strategies to mitigate cardiovascular disease risk in patients with DKD are important to improve outcomes in this patient²⁹ and are thus emphasized within the ADA and KDIGO holistic approach for improving outcomes in all patients with DKD (Fig. 41.13).⁵⁶³ Accordingly, ADA and KDIGO recommend moderate- or high-intensity statin (Table 41.10) as a component of first-line therapy in all patients with DKD, as appropriate per individual risk evaluation and treatment goals.^{563,641} As illustrated in Fig. 41.13, intensification of the antihyperlipidemic strategy (e.g., addition of ezetimibe, PCSK9 inhibitor, and icosapent ethyl), and initiation of antiplatelet therapies in the setting of clinical cardiovascular disease are recommended on the basis of atherosclerotic cardiovascular disease risk and response to first-line statin.^{563,641}

Therapeutic Approaches Under Study

While a study is under way to investigate finerenone's potential benefit as add-on to background SGLT2 inhibitor therapy (ClinicalTrials.gov Identifier: NCT05254002), early exploratory findings suggest finerenone may provide complementary risk reduction when used in combination with SGLT2 inhibitors and GLP-1 receptor agonists.^{642,643} In a phase II trial, an endothelial A receptor antagonist zibotentan was tested in combination with dapagliflozin compared with dapagliflozin alone in patients with or without type 2 diabetes and CKD (ZENITH-CKD; NCT04724837). After 12 weeks of treatment, the reduction in albuminuria for combination therapy was greater than with dapagliflozin alone. The main side effect was fluid retention, which was attenuated with dapagliflozin in the lower zibotentan dose group. This strategy will be tested in a large phase II clinical trial to follow in patients with or without type 2 diabetes and CKD characterized by high-grade albuminuria (UACR >700 mg/g).⁶⁴⁴ Study of Tirzepatide (LY3298176) in Participants With Overweight or Obesity and Chronic Kidney Disease With or Without Type 2 Diabetes (TREASURE-CKD; NCT05536804) is an ongoing 52-week trial exploring the effect of tirzepatide

on kidney oxygenation by MRI in addition to multiple other outcomes (e.g., eGFR and UACR change). Another ongoing study, “Semaglutide Cardiovascular Outcomes Trial in Patients With Type 2 Diabetes trial” (SOUL trial, NCT03914326), is examining the efficacy of oral semaglutide for combined cardiovascular and kidney outcomes among individuals with type 2 diabetes and established cardiovascular disease or CKD. The Renal Mode of Action of Semaglutide in Patients With Type 2 Diabetes and Chronic Kidney Disease (REMODEL trial, NCT04865770) is examining the kidney mechanism of action for semaglutide with MRI and kidney biopsies. A Phase II Randomized Controlled Study of Renal Autologous Cell Therapy (REACT) in Subjects With Type 2 Diabetes and Chronic Kidney Disease (REGEN-006; NCT05099770) is assessing the safety and efficacy of autologous renal cell therapy in patients with type 2 diabetes and CKD. A phase II clinical trial of aldosterone synthase inhibition, on top of the SGLT2 inhibitor empagliflozin and renin-angiotensin system inhibition, dose dependently reduced albuminuria over 14 weeks in patients with or without type 2 diabetes and CKD. Aldosterone synthase inhibition and SGLT2 inhibition together showed additive antialbuminuric efficacy that may translate into greater kidney protection. A phase II trial of 11,000 patients, EASI-KIDNEY, will subsequently test the aldosterone synthase inhibitor with SGLT2i inhibition in diabetic and nondiabetic participants with CKD for clinical kidney and cardiovascular endpoints.

Unmet Need: Heart and Kidney Risk Reduction in Type 1 Diabetes

While numerous agents have received recent approval to mitigate heart and kidney risk in the setting of type 2 diabetes and CKD, an area of large unmet need is therapeutic advancement for people with type 1 diabetes and CKD.⁶⁴⁵ In the setting of type 1 diabetes and CKD, current evidence-based approaches to slowing CKD progression are limited to renin-angiotensin system inhibitor therapy and optimization of glycemic and blood pressure control.^{32,563} While studies have been conducted with SGLT2 inhibitors in the setting of type 1 diabetes,⁶⁴⁶⁻⁶⁵³ agents from the class have not yet gained approval in this population due to risk for severe hypoglycemia and ketoacidosis. Nonetheless, analyses of hemodynamic effects of SGLT2 inhibitors in type 1 diabetes demonstrated comparable changes in eGFR and UACR as observed in type 2 diabetes studies.^{654,655} GLP-1RAs have similarly been studied for glycemic control and weight loss in type 1 diabetes, but their long-term effects on kidney and cardiovascular risk in type 1 diabetes have not been evaluated. Fortunately, three upcoming trials will study new therapies in people with type 1 diabetes and CKD. The REMODEL-T1D trial will evaluate the efficacy and safety of semaglutide for albuminuria, as well as kidney oxygenation and fibrosis by MRI.⁶⁵⁶ A Study to Learn How Well the Study Treatment Finerenone Works and How Safe It Is in People With Long-term Decrease in the Kidneys' Ability to Work Properly (Chronic Kidney Disease) Together With Type 1 Diabetes (FINE-ONE, NCT05901831) will additionally test the efficacy and safety of finerenone for albuminuria reduction as a “bridging biomarker” to the large outcome trials conducted in type 2 diabetes and CKD.^{624,625,657} And the effectiveness and safety of sotagliflozin in slowing

kidney function decline in persons with type 1 diabetes and moderate-to-severe diabetic kidney disease: The SUGAR-N-SALT Trial will test an SGLT2 inhibitor versus placebo for preservation of eGFR.

FUTURE DIRECTIONS

On the clinical side, recognition of interconnections of metabolic risk factors (e.g., obesity and diabetes) with cardiovascular complications and CKD has led to the concept of CKM syndrome. A new CKM initiative and risk calculator including eGFR and UACR, as well as conventional cardiovascular risk factors, has been developed by the American Heart Association.^{561,562} The focus of the CKM initiative includes screening, prevention, and holistic management as a systemic syndrome across a continuum of cardiovascular, kidney, and metabolic risks. On the discovery science side, incorporation of transcriptomic, proteomic, metabolomic, and lipidomic technologies, as well as use of artificial intelligence is being used by number of consortia including the Kidney Precision Medicine Project, Transformative Research in Diabetic Nephropathy (TRIDENT), and Biomarker Enterprise to Attack Diabetic Kidney Disease (BEAt-DKD). These approaches offer enhanced understanding of molecular mechanisms and phenotypes of DKD that may ultimately allow development and implementation of better-targeted biomarkers and therapies.^{308,658-660}

Clinical Relevance: Treatment

New therapies for the treatment of DKD have dramatically changed the standard of care in this population. SGLT-2 inhibitors are recommended in all patients with type 2 diabetes and CKD with an eGFR ≥ 20 mL/min/1.73 m² along with maximally tolerated RAS inhibitor therapy. Additional risk-based therapies can be considered to further reduce kidney and heart disease risks, including finerenone and/or GLP-1 receptor agonists. Multiple trials are in progress to further understand the incremental benefits of combination therapy in type 2 diabetes and DKD, as well as determine the role of these therapies in patients with type 1 diabetes.

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QUESTIONS

A. F. is a 48-year-old woman comes to you with uncontrolled type 1 diabetes, heart failure and new onset of severe albuminuria despite maximal dose of ramipril. She does not have diabetic retinopathy. Her urinary sediment is unremarkable and her urine ACR is 1600 mg/g. She has a normocytic anemia and normal platelet count. Her PCP send the patient to you for evaluation of DKD. Her lab work demonstrated a creatinine of 1.3 mg/dl. Which test is most likely to reveal the appropriate diagnosis?

- A. HgA1C
- B. A peripheral blood smear
- C. A Congo red staining on a kidney biopsy
- D. The presence of 2/3 urine collections with an ACR > 30 mg/g

Correct answer is C. Rationale: The presentation of this patient meets the criteria for a clinically indicated kidney biopsy. The presence of CHF, anemia, proteinuria and renal failure together with the absence of diabetic retinopathy makes the diagnosis of DKD highly unlikely and suggests amyloidosis.

B. N. is a 38-year-old man with a 15 years history of type 1 diabetes. His A1C is 7% and his blood pressure is 130/80. He comes to you for an upper respiratory infection that started 10 days prior. His Strep throat test is positive. You screen for DKD and find no albuminuria and normal eGFR. His complement is normal. He has microhematuria. A renal U/S demonstrates a complex mass that is confirmed by biopsy to be a renal cell carcinoma. When the patient undergoes nephrectomy, the pathologist reports thickening of the glomerular basement membrane in the non-tumoral portion of the parenchyma and no deposits. Immunofluorescence is negative.

The likely diagnosis is:

- A. IgA nephropathy
- B. Post-infectious GN
- C. Diabetic kidney disease
- D. Alport syndrome

Correct answer is C. Rationale: The presence of thickening of the GBM is a unique feature of DKD and is not a characteristic of any of the other diagnosis. IgA nephropathy would also be unlikely given the time from the upper respiratory infection. Alport would present with microhematuria but not with thickening of the GBM. Post-infection is not an option, as complement levels are normal and immunofluorescence is negative.

Which one of the following statements is true?

- A. Nodular glomerulosclerosis is pathognomonic for DKD
- B. In type 1 diabetic patients, interstitial fibrosis and tubular atrophy is an early finding and starts before diabetic glomerulopathy becomes established
- C. DKD lesions are more homogeneous in type 2 compared to type 1 diabetic patients
- D. There is substantial overlap between diabetic glomerular structural changes between normoalbuminuric and microalbuminuric patients

Correct answer: D. Rationale: There is substantial overlap between glomerular basement membrane

width and mesangial fractional volume values in normoalbuminuric and microalbuminuric type 1 diabetic patients. Nodular glomerulosclerosis is not pathognomonic for DKD and can be seen in other conditions, such as light chain deposition disease, amyloidosis, idiopathic nodular glomerulosclerosis, membranoproliferative glomerulonephritis, fibrillary and immunotactoid glomerulopathy, fibronectin glomerulopathy, collagen type III glomerulopathy, congenital cyanotic heart disease, and cystic fibrosis. DKD lesions are more heterogeneous in type 2 diabetic patients who develop diabetes later in life when they already have hypertension and atherosclerosis.

J. K. is a 71 yo woman with type 2 diabetes and stage 3a CKD. Her eGFR is 52 mL/min/1.73m² and her UACR is 60 mg/g. Her current medication regimen consists of metformin 500 mg twice daily, linagliptin 5 mg once daily, lisinopril 40 mg once daily, and atorvastatin 40 mg once daily. Her current HbA1c is 6.9%. Which of the following would be the **most appropriate** change(s) to JK's regimen at this time based on current guidelines?

- A. Discontinue metformin
- B. **Start a SGLT2 inhibitor**
- C. Start a GLP-1 receptor agonist
- D. Do nothing – She is currently at her HbA1c goal

Correct answer: B. Rationale: According to ADA/KDIGO recommendations, SGLT2 inhibitors are considered standard of care for patients with type 2 diabetes and CKD. Recommendations for use of a SGLT2 inhibitor to mitigate cardiorenal risk are independent of the need for additional glucose lowering. Initiation of a SGLT2 inhibitor would be indicated.

Question #5

BG is a 62 yo man with type 2 diabetes and CKD. His eGFR is 33 mL/min/1.73m² and his UACR is 300 mg/g. His current medication regimen consists of empagliflozin 10 mg once daily, metformin 500 mg twice daily, valsartan 80 mg once daily, and rosuvastatin 20 mg once daily. His current HbA1c is 8.2%. Which of the following additions would be preferentially recommended to help BG achieve his individualized glycemic goals and further mitigate his cardiovascular and kidney risk?

- A. Start a DPP-4 inhibitor
- B. Start finerenone
- C. **Start a GLP-1 receptor agonist**
- D. Start basal insulin

Correct answer: C. Rationale: A GLP-1 receptor agonist with evidence of cardiovascular benefit is preferentially recommended in patients with type 2 diabetes and CKD who do not meet their individualized glycemic goals despite first-line treatment with an SGLT2 inhibitor and metformin. A GLP-1 receptor agonist would additionally further mitigate heart and kidney risk. While starting finerenone would help mitigate his cardiovascular and kidney risk, finerenone does not have glucose-lowering effects. Conversely, a DPP-4 inhibitor or basal insulin could lower his HbA1c, but DPP-4 inhibitors and basal insulin products have not been shown to provide heart or kidney benefit in outcome trials.